Sensing-Aided Precoding With High-Dynamic Moving Scatterers

Luning Lin, *Graduate Student Member, IEEE*, Chengwei Zhou, *Senior Member, IEEE*, Hang Zheng, Zhiguo Shi, *Fellow, IEEE*, Sergiy A. Vorobyov, *Fellow, IEEE*, and Robert W. Heath Jr., *Fellow, IEEE*

***Abstract*—Sensing information can be leveraged to reduce the overheads associated with establishing and maintaining multiple-input and multiple-output (MIMO) communication links. Such information can be acquired from integrated sensing and communication (ISAC) capabilities. In this paper, we use sensing information to make precoding for spatial multiplexing more robust in high-dynamic environments with both quasi-static and mobile scattering objects. The significant variability of scattered paths requires frequent reconfigurations of the precoding matrices. To address this, we propose a sensing-aided precoding scheme for channels with high dynamics, leveraging the identification and localization of moving targets to concentrate energy mainly on slow-varying paths. Specifically, the geometric structure of the time-varying channel matrix is analyzed to uncover deviations in the location parameters of moving targets. A channel division criterion is then devised to partition the channel matrix into two matrices characterizing the paths of background scatterers and moving targets, respectively. Exploiting the digital-analog dual orthogonality between these matrices, precoding and combining matrices are jointly designed to suppress transmission leakage over mobile paths in both analog and digital domains. Numerical simulations validate that the proposed method achieves high spectral efficiency without requiring explicit channel tracking, thereby reducing complexity and overheads compared to conventional MIMO precoding methods.**

***Index Terms*—High-dynamic scenarios, integrated sensing and communication, precoding design, sensing-aided communication.**

Received 11 November 2024; revised 20 May 2025; accepted 4 July 2025. Date of publication 10 July 2025; date of current version 15 August 2025. The work of Luning Lin, Chengwei Zhou, Hang Zheng, and Zhiguo Shi was supported in part by the National Natural Science Foundation of China under Grant 62271444 and Grant U21A20456, in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LZ23F010007, and in part by the Research Project of the State Key Laboratory of Industrial Control Technology under Grant ICT2025A03. An earlier version of this paper was presented at the IEEE International Conference on Acoustics, Speech and Signal Processing, Seoul, South Korea, Apr. 2024 [DOI: 10.1109/ICASSP48485.2024.10448481]. The associate editor coordinating the review of this article and approving it for publication was Mohammed Nabil El Korso. *(Corresponding author: Chengwei Zhou.)*

Luning Lin, Chengwei Zhou, Hang Zheng, and Zhiguo Shi are with the College of Information Science and Electronic Engineering, Zhejiang University, Hangzhou 310027, China (e-mail: luninglin@zju.edu.cn; zhouchw@zju.edu.cn; hangzheng@zju.edu.cn; shizg@zju.edu.cn).

Sergiy A. Vorobyov is with the Department of Information and Communications Engineering, Aalto University, 00076 Aalto, Finland (e-mail: sergiy.vorobyov@aalto.fi).

Robert W. Heath Jr. is with the Department of Electrical and Computer Engineering, University of California San Diego, San Diego, CA 92093 USA (e-mail: rwheathjr@ucsd.edu).

Digital Object Identifier 10.1109/TSP.2025.3587089

I. Introduction

THE support for sensing appears to be a key feature for the sixth-generation (6G) mobile networks [2], [3]. By effectively integrating sensing functionalities, 6G can better enable applications including connected and automated vehicles, smart factories, and unmanned aerial vehicle (UAV) networks [4]. To achieve integrated sensing and communication (ISAC) within the mobile networks, auxiliary sensing information can be harvested from various sources, such as camera images [5], [6], [7], Global Positioning System [8], [9], radar signals [10], [11], [12], [13], [14], and LiDAR point clouds [15], [16], [17], [18]. Apart from imposing additional sensors and equipments, a cost-effective implementation of sensing-aided communication systems can be realized through modifications to existing communication infrastructures [19], [20]. Leveraging advanced hardware and versatile processing capabilities, these infrastructures can exploit transmitted waveforms to extract sensing information from non-cooperative targets via radar signal processing [21], [22]. The sensing information can then be leveraged in multiple-input and multiple-output (MIMO) precoding designs to enhance the efficiency of communication links.

The exploitation of sensing information in precoding design differs between single-path and multi-path channel conditions. Under single-path channels, the angle-of-arrival (AOA) and angle-of-departure (AOD) obtained from sensing procedure are employed to assist the formation of pencil-like steering beams between the base station (BS) and the mobile station (MS) [12], [23]. The predicted angular parameters can facilitate consistent updates of beam direction for improved data transmission towards a moving MS [24]. Additionally, the spatial correlation can also be used to facilitate beam configuration [25]. These precoding designs were typically tailored to reducing the overheads associated with brute-force beam training for single-stream communications. Thus, only the strongest propagation path is utilized while other paths are ignored or suppressed, thereby limiting the achievable rate in the presence of multiple scatterers.

Under multi-path channels, location information of scatterers can be utilized in precoding design to augment spatial multiplexing capabilities for multi-stream data transmission. For example, accurate estimates of location parameters for scatterers have shown to enhance channel estimation accuracy and facilitate higher data rates in multi-stream transmission [26], [27]. The spatial covariance of radar signals has also been

1053-587X © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

exploited to establish communication links without requiring the transmitted waveform as a reference [10]. Meanwhile, the relationships between radar and communication channels have been revealed using deep learning techniques to configure the communication links [14]. While these studies have demonstrated the potential of sensing-assisted communication for multi-stream transmission, they overlooked the dynamic nature of scatterers and their mobility-induced impact on transmission performance. Consequently, these efforts have focused primarily on initial beam alignment, with limited attention given to maintaining or adaptively updating the precoding strategy based on predicted time-varying channel state information (CSI) in high-dynamic environments. Even though exploiting other sensing modalities offers potential performance gains [28], [29], [30], [31], [32], these multisensing approaches also do not address the challenges associated with mobility in the environment.

In dynamic environments with high-mobility scatterers such as vehicles and UAVs, the communication CSI varies significantly. To maintain reliable links, a straightforward approach is frequent CSI tracking, which involves increased allocation of training symbols and feedback resources. Given the finite budget of time-frequency resources, allocating more resources for channel estimation decreases the achievable rate because fewer resources are available for data transmission. To mitigate this tradeoff, a simultaneous data and pilot transmission strategy has been proposed [33], [34], where the training sequence is transmitted in the null space of the channel matrix. Nevertheless, introducing additional pilots leads to an increase in both energy consumption and algorithmic complexity. Modulation schemes in the delay-Doppler domain characterize time-varying channels via the delay and Doppler of each propagation path [35], [36]. This representation yields a sparser channel structure which evolves at a slower rate over time compared to that in the time-frequency domain. However, frequent channel estimation remains necessary, as the complex path gains still change rapidly. Hence, there is a significant opportunity to exploit sensing information to reduce the overheads for precoded MIMO in dynamic channels.

In this paper, we propose a sensing-aided precoding method for dynamic MIMO channels. The key idea is to leverage sensing information, obtained from bistatic target localization within an ISAC framework, to identify the propagation paths associated with moving targets and enhance precoding design for multi-stream transmission. In particular, the targets identified and localized in radar signal processing are in fact corresponding to mobile scatterers in the environment. Then, the channel can be decomposed into two constituent components: one associated with quasi-static paths and one associated with mobile paths. In this way, the precoding can be designed to focus energy on the quasi-static communication paths, thereby improving robustness despite mobility of the surrounding scatterers.

Our technical contributions are summarized as follows.

- We provide a geometrical analysis of the time-varying channel structure within a time frame, revealing the channel deviation attributed to the changing locations of scattering objects. This enables a more realistic representation of time-varying channel modeling in high-dynamic environments with both quasi-static and mobile scatterers.
- We propose a sensing-aided precoding design for high-dynamic environments with moving scatterers. By estimating the positions and velocities of scatterers, the corresponding propagation paths are classified into mobile and quasi-static categories. This classification is then incorporated into a sensing-aided hybrid precoding design, where energy leakage over high-mobility paths is suppressed in both analog and digital domains.
- We evaluate the performance of the proposed method in terms of achievable rates and computational complexities, and compare it with state-of-the-art precoding methods. Simulation results demonstrate that the proposed method maintains high spectral efficiency over an entire time frame without requiring a separate channel tracking procedure, leading to reduced computational complexity.

Some preliminary results were presented in the conference paper [1], which focused on channel estimation with a fully-digital system. In this paper, we design sensing-aided precoding tailored for hybrid analog-digital architectures, which circumvents the excessive hardware cost and power consumption of fully-digital systems. Moreover, we provide extensive theoretical analysis, detailing the impact of high-mobility scatterers on channel dynamics. Comprehensive simulation validates our method.

The rest of the paper is organized as follows. In Section II, we present the deployed system model. Section III introduces the identification and localization of scatterers. In Section IV, we analyze the performance degradation of conventional precoding designs and detail the proposed precoding strategy. Section V investigates the impact of low-resolution digital-to-analog converters (DACs) on the performance of the proposed method. In Section VI, we analyze the scatterer identifiability and computational complexity of the proposed method. Section VII presents the simulation results, and Section VIII concludes the paper and discusses potential future directions.

We make use of the notations summarized in Table I.

II. System Model

A. System and Frame Structure

We present the downlink signal models for a hybrid MIMO system operating within the millimeter wave (mmWave) frequency band and introduce the high-dynamic channel model associated with the motion of scatterers, as shown in Fig. 1. We adopt a two-stage frame structure with a light level of integration between sensing and communication, as shown in Fig. 2. This design requires minimal modifications to existing communication protocols and reduces the resource tradeoffs encountered in ISAC systems with signal-level integration.

Each time frame consists of G subframes. Within a time frame, the first subframe is dedicated to transmitting frequency modulated continuous wave (FMCW) chirps as a preamble

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

TABLE I
LIST OF NOTATION

Symbol	Description
$a, \boldsymbol{a}, \boldsymbol{A}, \boldsymbol{\mathcal{A}}$	Scalar, vector, matrix, and tensor
$\boldsymbol{a}_{(i)}$	The i-th element of vector $\boldsymbol{a}$
$[\boldsymbol{A}]_{(i,j)}$	The (i,j)-th element of matrix $\boldsymbol{A}$
$[\boldsymbol{A}]_{(\cdot,j)}$	The j-th column of matrix $\boldsymbol{A}$
$[\boldsymbol{A}]_{(i_1:i_2,j_1:j_2)}$	The submatrix corresponding to the i_1-th to i_2-th rows and the j_1-th to j_2-th columns of matrix $\boldsymbol{A}$
$[\boldsymbol{\mathcal{A}}]_j$	Mode-j unfolding of tensor $\boldsymbol{\mathcal{A}}$
$\mathbb{C}$, $\mathbb{N}$	Complex and natural number sets
j	Imaginary unit
$(\cdot)^{\mathrm{T}}, (\cdot)^{\mathrm{H}}, (\cdot)^{\dagger}$	Transpose, Hermitian transpose, and pseudoinverse
$\circ, \odot$	Outer product and Hadamard product
$\|\cdot\|_{\mathrm{F}}, \|\cdot\|_2$	Frobenius norm and Euclidean norm
$[\![\cdot]\!]$	Canonical polyadic decomposition of a tensor
$\lvert\cdot\rvert$	Absolute value or matrix determinant
$\kappa(\cdot)$	Kruskal's rank of a matrix
$\mathrm{sgn}(\cdot)$	Sign function
$\mathrm{diag}(\cdot)$	Diagonal operator
$\Re(\cdot), \Im(\cdot)$	Real/imaginary part of a complex number
$\max(\cdot), \min(\cdot)$	Maximum operator and minimum operator
$\boldsymbol{I}$, $\boldsymbol{0}$	Identity matrix and zero matrix

for parameter estimation,[1] supporting both channel estimation and target localization. These FMCW chirps are generated via a dedicated hardware unit without using digital precoders and DACs (see Fig. 1). Simultaneously, the MS serves as a bistatic radar receiver. This dual MS functionality is realized by leveraging its standard communication hardware and software. Specifically, hardware feasibility stems from operating sensing and communication functions within the same frequency band, enabling the reuse of existing RF front-ends at the MS. Furthermore, the programmability of modern MS baseband modules facilitates efficient implementation of necessary radar signal processing algorithms. The feasibility of embedding sensing functionality within MSs is well supported by existing prototypes [38].

The estimated parameters are fed back to the BS at the end of the first subframe. This feedback includes parameters from a limited number of scatterers [39]. We assume the feedback transmission completes before the second subframe, utilizing a guard interval for assurance. In the remaining $(G-1)$ subframes, the BS and MS compute their respective precoding and combining matrices within each subframe, enabling data transmission based on orthogonal frequency-division multiplexing (OFDM) signaling. These matrices are derived using the estimated parameters from the first stage. A higher estimation accuracy of these parameters leads to higher achievable rate in data transmission.

We make several assumptions about the high-dynamic environment as follows:

1) Downlink signals reflected by moving targets (e.g., UAVs, vehicles) contribute to the channel observed at the MS. These moving targets are treated as point scatterers in the channel model, which is a typical setting in relevant literature of perceptive mobile networks [40].
2) The location parameters of scatterers, including AOA, AOD and distance, remain constant within a subframe but change across different subframes [41]. This assumption is reasonable given the short subframe duration. For example, a 1 ms subframe limites the maximum movement of a 30 m/s target to just 0.03 m.
3) The scatterers' motion trajectories are approximated as linear within one time frame, owing to the short channel estimation interval and their low acceleration. This approximation is justified by the negligible lateral deviation over a frame duration. For instance, a target moving at 30 m/s on a 100 m radius curved path deviates laterally by only about 7.2 mm over a 40 ms frame.

B. Signal Model at the Channel Estimation Stage

At the channel estimation stage, the BS transmits a sequence of M FMCW chirps for bistatic radar processing to achieve downlink channel estimation and moving target localization at the MS. We denote the FMCW pulse repetition interval (PRI) as T_{r}, the chirp duration as T_0 ($T_0 \leq T_{\mathrm{r}}$), the signal bandwidth as B, and the chirp rate as $\gamma = B/T_0$. To illuminate potential targets in the environment, a directional scanning beam is generated using the beamforming vector $\tilde{\boldsymbol{f}}_{\mathrm{RF},\tilde{m}} = \boldsymbol{a}_{\mathrm{BS}}(\vartheta_{\tilde{m}}) \in \mathbb{C}^{N_{\mathrm{BS}}}$ corresponding to the $\tilde{m}$-th chirp, $\tilde{m} = 1, 2, \cdots, M$. Here, $\boldsymbol{a}_{\mathrm{BS}}(\vartheta_{\tilde{m}})$ denotes the array response vector at the BS for the steering angle $\vartheta_{\tilde{m}}$, given by

$$\boldsymbol{a}_{\mathrm{BS}}(\vartheta_{\tilde{m}}) = \frac{1}{\sqrt{N_{\mathrm{BS}}}} \left[1, e^{-\mathrm{j}\pi \sin\vartheta_{\tilde{m}}}, \cdots, e^{-\mathrm{j}\pi(N_{\mathrm{BS}}-1)\sin\vartheta_{\tilde{m}}}\right]^{\mathrm{T}}. \tag{1}$$

The set of steering angles $\{\vartheta_{\tilde{m}}\}_{\tilde{m}=1}^{M}$ is selected based on a steering grid designed to ensure sufficient coverage of the sensing region. Then, the normalized transmitted signal vector in the $\tilde{m}$-th chirp can be expressed as

$$\boldsymbol{s}_{\tilde{m}}(t) = e^{\mathrm{j}2\pi(f_c t + 0.5\gamma t^2)} \tilde{\boldsymbol{f}}_{\mathrm{RF},\tilde{m}}, \quad t \in [\tilde{m}T_{\mathrm{r}}, \tilde{m}T_{\mathrm{r}} + T_0]. \tag{2}$$

At the MS side, the received signals are first combined with the analog combining matrix $\tilde{\boldsymbol{W}}_{\mathrm{RF}} \in \mathbb{C}^{N_{\mathrm{MS}} \times M_{\mathrm{MS}}}$, and then sampled by analog-to-digital converter (ADCs) at the rate of f_{s}. The digitized signals are subsequently cross-correlated with the transmitted chirp signals and processed by the digital combining matrix $\tilde{\boldsymbol{W}}_{\mathrm{BB}} \in \mathbb{C}^{M_{\mathrm{MS}} \times M_{\mathrm{MS}}}$.

Assuming that there is no line-of-sight (LOS) paths between the BS and the MS, and the propagation paths correspond to L scattering objects. The path of the ℓ-th ($\ell = 1, 2, \cdots, L$) scatterer is characterized by complex path gain β_ℓ, propagation delay $\tau_{\ell,1}$, Doppler frequency $f_{\ell,1}$, AOA $\theta_{\ell,1}$, and AOD $\phi_{\ell,1}$ within the first subframe. Then, the discrete received signal

[1]FMCW chirps are employed for channel estimation instead of OFDM pilots due to their robustness against large Doppler shifts, as well as their hardware and power efficiency provided by a low peak-to-average power ratio (PAPR). Specifically, high PAPR can arise in frequency-flat beam training schemes used in OFDM signaling [37]. These schemes use the same precoders and pilots across all subcarriers, resulting in the coherent combination of signals in the time domain and creating large power peaks.

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

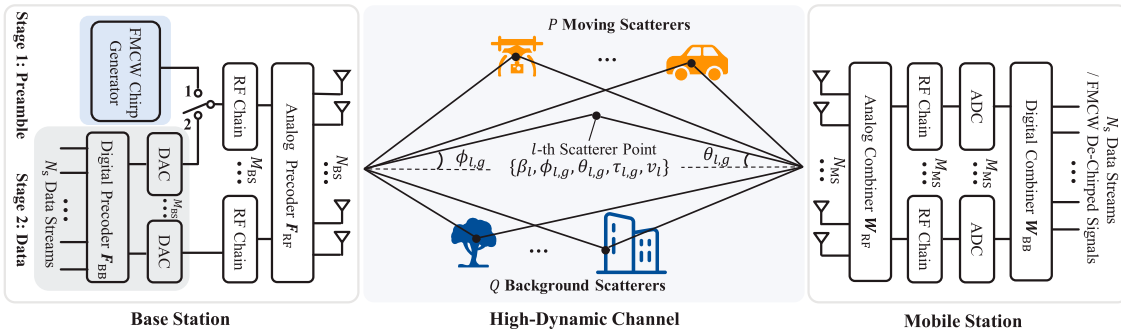


Fig. 1. Illustration of the deployed system and the high-dynamic channel. The BS is equipped with a uniform linear array (ULA) of $N_{\rm BS}$ antennas and $M_{\rm BS}$ radio frequency (RF) chains, while the MS is equipped with a ULA of $N_{\rm MS}$ antennas and $M_{\rm MS}$ RF chains. The inter-element spacing d of the ULAs is set to half of the signal wavelength $\lambda = c/f_c$, where c denotes the light speed, and f_c denotes the carrier frequency.

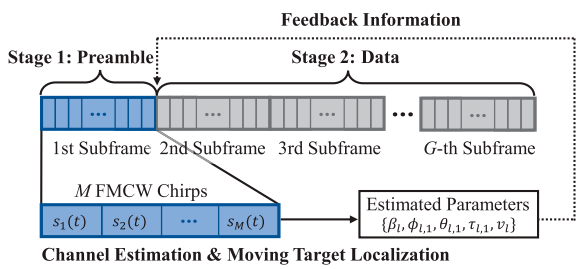


Fig. 2. Illustration of the frame structure.

vector corresponding to the $\tilde{k}$-th ($\tilde{k} = 1, 2, \cdots, K$) fast-time sample within one PRI can be represented as

$$\boldsymbol{r}_{\tilde{k}}(\tilde{m}) = \sum_{\ell=1}^{L} \beta_\ell e^{-\mathrm{j}2\pi f_c \tau_{\ell,1}} e^{\mathrm{j}2\pi\gamma\left(-\tau_{\ell,1}+\frac{f_{\ell,1}}{f_c}\tilde{m}T_{\rm r}\right)\frac{\tilde{k}}{f_{\rm s}}} e^{\mathrm{j}2\pi f_{\ell,1}\tilde{m}T_{\rm r}} \times \tilde{\boldsymbol{W}}_{\rm BB}^{\rm H}\tilde{\boldsymbol{W}}_{\rm RF}^{\rm H}\boldsymbol{a}_{\rm MS}(\theta_{\ell,1})\tilde{\boldsymbol{f}}_{{\rm RF},\tilde{m}}^{\rm H}\boldsymbol{a}_{\rm BS}(\phi_{\ell,1}) + \boldsymbol{z}_{\tilde{k}}(\tilde{m}). \tag{3}$$

Here, $\boldsymbol{a}_{\rm MS}(\theta_{\ell,1}) = \frac{1}{\sqrt{N_{\rm MS}}}[1, e^{-\mathrm{j}\pi\sin\theta_{\ell,1}}, \cdots, e^{-\mathrm{j}\pi(N_{\rm MS}-1)\sin\theta_{\ell,1}}]^{\rm T}$ is the array response vector at the MS, $\boldsymbol{z}_{\tilde{k}}(\tilde{m}) \sim \mathcal{CN}\left(\boldsymbol{0}, \sigma_{\rm n}^2\boldsymbol{I}\right)$ represents the circularly symmetric complex white Gaussian noise with zero mean and variance $\sigma_{\rm n}^2$. Since $f_{\ell,1}/f_c \ll 1$ typically holds true [42], the range-Doppler coupling term $e^{\mathrm{j}2\pi\gamma(\frac{f_{\ell,1}}{f_c}\tilde{m}T_{\rm r})\frac{\tilde{k}}{f_{\rm s}}}$ in (3) can be neglected, and $\boldsymbol{r}_{\tilde{k}}(\tilde{m})$ can be approximated by

$$\boldsymbol{r}_{\tilde{k}}(\tilde{m}) \approx \sum_{\ell=1}^{L} \tilde{\beta}_\ell e^{-\mathrm{j}2\pi\gamma\tau_{\ell,1}\frac{\tilde{k}}{f_{\rm s}}} e^{\mathrm{j}2\pi f_{\ell,1}\tilde{m}T_{\rm r}} \tilde{\boldsymbol{a}}_{\rm MS}(\theta_{\ell,1}) \times \tilde{\boldsymbol{f}}_{{\rm RF},\tilde{m}}^{\rm H}\boldsymbol{a}_{\rm BS}(\phi_{\ell,1}) + \boldsymbol{z}_{\tilde{k}}(\tilde{m}), \tag{4}$$

where $\tilde{\beta}_\ell = \beta_\ell e^{-\mathrm{j}2\pi f_c\tau_{\ell,1}}$ denotes the equivalent complex path gain, and $\tilde{\boldsymbol{a}}_{\rm MS}(\theta_{\ell,1}) = \tilde{\boldsymbol{W}}_{\rm BB}^{\rm H}\tilde{\boldsymbol{W}}_{\rm RF}^{\rm H}\boldsymbol{a}_{\rm MS}(\theta_{\ell,1})$ is the equivalent steering vector after combining at the MS.

C. Signal Model at the Data Transmission Stage

At the data transmission stage, the BS communicates $N_{\rm s}(N_{\rm s} \le M_{\rm MS} \le M_{\rm BS})$ independent data streams to the MS using OFDM signals. For each data stream, K orthogonal subcarriers are employed, and the number of OFDM symbols within each subframe is M. We denote the analog and digital precoding matrices as $\boldsymbol{F}_{\rm RF} \in \mathbb{C}^{N_{\rm BS}\times M_{\rm BS}}$ and $\boldsymbol{F}_{{\rm BB},k,g}(m) \in \mathbb{C}^{M_{\rm BS}\times N_{\rm s}}$, respectively. Let $\dot{\boldsymbol{x}}_{k,g}(m) \in \mathbb{C}^{N_{\rm s}}$ be the transmitted symbol sequence of quadrature phase-shift keying (QPSK) constellation, the precoded transmitted signal vector of the k-th subcarrier and the m-th transmitted OFDM symbol of the g-th subframe can be represented as

$$\boldsymbol{x}_{k,g}(m) = \boldsymbol{F}_{\rm RF}\boldsymbol{F}_{{\rm BB},k,g}(m)\dot{\boldsymbol{x}}_{k,g}(m), \quad \forall k = 1, 2, \cdots, K, m = 1, 2, \cdots, M, g = 2, 3, \cdots, G. \tag{5}$$

At the MS, the received signals are combined by the analog combining matrix $\boldsymbol{W}_{\rm RF} \in \mathbb{C}^{N_{\rm MS}\times M_{\rm MS}}$ and digital combining matrix $\boldsymbol{W}_{{\rm BB},k,g}(m) \in \mathbb{C}^{M_{\rm MS}\times N_{\rm s}}$, respectively. The received signal vector after hybrid combining can be represented as

$$\boldsymbol{y}_{k,g}(m) = \boldsymbol{W}_{{\rm BB},k,g}^{\rm H}(m)\boldsymbol{W}_{\rm RF}^{\rm H}\left[\boldsymbol{H}_{k,g}(m)\boldsymbol{x}_{k,g}(m) + \boldsymbol{n}_{k,g}(m)\right], \tag{6}$$

where $\boldsymbol{H}_{k,g}(m) \in \mathbb{C}^{N_{\rm MS}\times N_{\rm BS}}$ denotes the time-varying channel matrix, and $\boldsymbol{n}_{k,g}(m) \sim \mathcal{CN}\left(\boldsymbol{0}, \sigma_{\rm n}^2\boldsymbol{I}\right)$ is the noise term. Note that, since the analog precoding and combining are implemented by phase shifters before digitization, uniform phase shifts are applied to signals of different subcarriers. In contrast, the digital precoding matrix $\boldsymbol{F}_{{\rm BB},k,g}(m)$ and combining matrix $\boldsymbol{W}_{{\rm BB},k,g}(m)$ vary across subcarriers and can be flexibly changed for each symbol duration.

D. Dynamic Channel Model

In the following, we characterize the dynamic channel matrix and its parametric model, with a focus on how scatterer motion induces geometric variations in propagation paths. In particular, each path corresponds to either a quasi-static background scatterer or a moving target. Among all the propagation paths, P paths correspond to the moving targets with high velocity $\bar{v}_p, p = 1, 2, \cdots, P$, while the remaining $Q = L - P$ paths correspond to the background scatterers with low velocity $\bar{v}_q, q = P+1, P+2, \cdots, L$, satisfying $|\bar{v}_p| \gg |\bar{v}_q|$.[2] The motion of

[2]Scatterers are classified as mobile or quasi-static based on a velocity threshold given by 3GPP specifications [43] (e.g., $\le$ 3 km/h for typical urban background scatterers). Since both parameter estimation and precoding designs use threshold-independent methods, the choice of this threshold does not affect estimation and precoding performance.

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

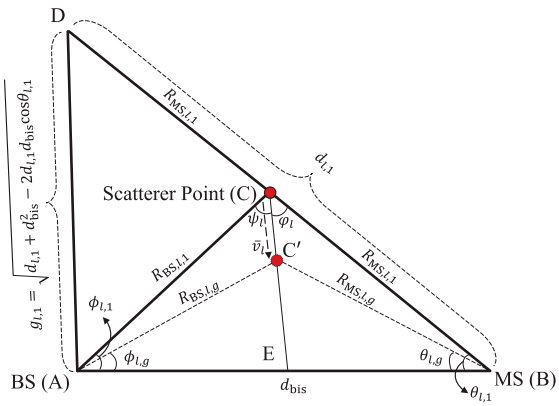


Fig. 3. Illustration of the geometric relationships among channel parameters for a scatterer. The locations of the ℓ-th scatterer, the BS, and the MS are denoted by points C, A, B, respectively. We denote the angle between line segments CA and CE as ψ_ℓ, and the angle between line segments CB and CE as φ_ℓ. Over a time frame, the scatterer at point C moves along the line CE with velocity $\bar{v}_\ell$.

each scatterer induces changes in the location parameters, including angles and distances, across different subframes.

To demonstrate such changes, the geometrical structure of the high-dynamic scenario is represented using a triangular topology, as shown in Fig. 3. We assume that the scatterer moves along a straight line within the duration of a frame, as justified in the third assumption at the end of Subsection II-A. Considering that the scatterer point reaches point C′ by the end of the g-th subframe, the total distance $d_{\mathrm{CC}'}$ of movement is calculated as

$$d_{\mathrm{CC}'} = (g-1)MT_{\mathrm{s}}\bar{v}_\ell, \tag{7}$$

where T_{s} denotes the OFDM symbol duration. The distance between point C′ to the MS and BS can then be calculated as

$$\begin{aligned} R_{\mathrm{MS},\ell,g} &= \sqrt{R_{\mathrm{MS},\ell,1}^2 + d_{\mathrm{CC}'}^2 - 2R_{\mathrm{MS},\ell,1}d_{\mathrm{CC}'}\cos\varphi_\ell}, \\ R_{\mathrm{BS},\ell,g} &= \sqrt{R_{\mathrm{BS},\ell,1}^2 + d_{\mathrm{CC}'}^2 - 2R_{\mathrm{BS},\ell,1}d_{\mathrm{CC}'}\cos\psi_\ell}, \end{aligned} \tag{8}$$

where

$$\begin{aligned} \varphi_\ell &= (\pi - |\theta_{\ell,1}| - |\phi_{\ell,1}|)/2 + \delta_\ell, \\ \psi_\ell &= (\pi - |\theta_{\ell,1}| - |\phi_{\ell,1}|)/2 - \delta_\ell, \end{aligned} \tag{9}$$

and δ_ℓ denotes the unknown angle between the bistatic velocity and the direction of movement. As auxiliary parameters, the lengths of line segments CE, C′E, AE, and BE are calculated as $d_{\mathrm{CE}} = R_{\mathrm{MS},\ell,1}\sin|\theta_{\ell,1}|/\sin(\pi - |\theta_{\ell,1}| - |\varphi_\ell|)$, $d_{\mathrm{C'E}} = d_{\mathrm{CE}} - d_{\mathrm{CC}'}$, $d_{\mathrm{AE}} = d_{\mathrm{CE}}\sin|\psi_\ell|/\sin|\phi_{\ell,1}|$, and $d_{\mathrm{BE}} = d_{\mathrm{CE}}\sin|\varphi_\ell|/\sin|\theta_{\ell,1}|$, respectively. Then, the location parameters AOA $\theta_{\ell,g}$, AOD $\phi_{\ell,g}$, and length of propagation path $d_{\ell,g}$ are respectively represented as

$$\begin{aligned} \theta_{\ell,g} &= \arccos\left(\frac{R_{\mathrm{MS},\ell,g}^2 + d_{\mathrm{BE}}^2 - d_{\mathrm{C'E}}^2}{2R_{\mathrm{MS},\ell,g}d_{\mathrm{BE}}}\right), \\ \phi_{\ell,g} &= \arccos\left(\frac{R_{\mathrm{BS},\ell,g}^2 + d_{\mathrm{AE}}^2 - d_{\mathrm{C'E}}^2}{2R_{\mathrm{BS},\ell,g}d_{\mathrm{AE}}}\right), \\ d_{\ell,g} &= R_{\mathrm{BS},\ell,g} + R_{\mathrm{MS},\ell,g}. \end{aligned} \tag{10}$$

Accordingly, the propagation delay and the Doppler frequency of the ℓ-th scatterer in the g-th subframe can be represented as

$$\tau_{\ell,g} = \left(R_{\mathrm{MS},\ell,g} + R_{\mathrm{BS},\ell,g}\right)/c \tag{11}$$

and

$$f_{\ell,g} = \frac{1}{\lambda}\frac{\mathrm{d}\left[R_{\mathrm{MS},\ell,g} + R_{\mathrm{BS},\ell,g}\right]}{\mathrm{d}t} = \frac{2v_\ell}{\lambda}\sin\frac{(|\phi_{\ell,g}| + |\theta_{\ell,g}|)}{2}, \tag{12}$$

respectively. Here, $v_\ell = \bar{v}_\ell\cos\delta_\ell$ denotes the bistatic velocity. The Doppler shift $f_{\ell,g}$ in (12) reveals only the radial projection of the actual velocity $\bar{v}_\ell$ along the bistatic path, leaving the full motion direction and speed ambiguous. Without additional knowledge, only the bistatic velocity v_ℓ can be inferred under the current system setting, making it infeasible to predict the future positions of the scatterers.

Then, the time-varying channel matrix corresponding to the m-th symbol of the g-th subframe is formulated as

$$\begin{aligned} \boldsymbol{H}_{k,g}(m) = &\sum_{\ell=1}^{L}\beta_\ell e^{\mathrm{j}2\pi\left(mT_{\mathrm{s}}f_{\ell,g} + \sum_{\bar{g}=2}^{g-1}MT_{\mathrm{s}}f_{\ell,\bar{g}} + MT_{\mathrm{r}}f_{\ell,1}\right)} \\ &\times e^{-\mathrm{j}2\pi k\tau_{\ell,g}\Delta}\boldsymbol{a}_{\mathrm{MS}}\left(\theta_{\ell,g}\right)\boldsymbol{a}_{\mathrm{BS}}^{\mathrm{H}}\left(\phi_{\ell,g}\right), \end{aligned} \tag{13}$$

where Δ denotes the subcarrier spacing. Note that, the location parameters $\theta_{\ell,g}$, $\phi_{\ell,g}$, and $d_{\ell,g}$ in (10) change across different subframes, resulting in variations in the channel matrices $\boldsymbol{H}_{k,g}(m)$ corresponding to various subframes.

III. Moving Target Identification and Localization

To enable reliable data transmission in dynamic channels, it is important to identify the parameters of the moving targets and background scatterers to facilitate subsequent precoding design. In this regard, the moving targets can be distinguished and localized through a joint estimation of multiple parameters associated with each propagation path.

A. Tensor Formulation for Auto-Paired Parameter Estimation

By concatenating $\boldsymbol{r}_{\tilde{k}}(\tilde{m})$ in (4) across K fast-time samples and M chirps, the received signals over the duration of the first subframe is represented in the form of a third-order tensor with elements given as

$$[\boldsymbol{\mathcal{Y}}_1]_{(m_{\mathrm{MS}},\tilde{k},\tilde{m})} = [\boldsymbol{r}_{\tilde{k}}(\tilde{m})]_{(m_{\mathrm{MS}})},\ m_{\mathrm{MS}} = 1, 2, \cdots, M_{\mathrm{MS}}. \tag{14}$$

The low-rank structure of the tensor $\boldsymbol{\mathcal{Y}}_1$, which stems from the spatial sparsity of the propagation channel, enables canonical polyadic (CP) decomposition. This facilitates joint estimation of path-specific parameters with automatic pairing [44], [45]. Moreover, it reduces computational complexity compared to multi-dimensional search by allowing for one-dimensional search-based or closed-form solutions for individual parameters.

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

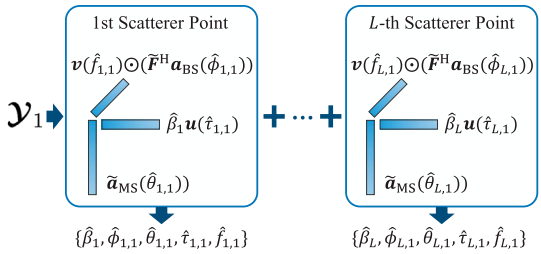


Fig. 4. Illustration of CP decomposition of received signal tensor $\boldsymbol{\mathcal{Y}}_1$.

Specifically, by substituting the expression of $\boldsymbol{r}_{\tilde{k}}(\tilde{m})$ in (4) into (14), $\boldsymbol{\mathcal{Y}}_1$ can be formulated as the following rank-L CP model

$$\boldsymbol{\mathcal{Y}}_1=\sum_{\ell=1}^{L}\tilde{\boldsymbol{a}}_{\mathrm{MS}}(\theta_{\ell,1})\circ\left[\tilde{\beta}_\ell\boldsymbol{u}(\tau_{\ell,1})\right]\circ\left[\boldsymbol{v}(f_{\ell,1})\right.$$
$$\left.\odot\left(\tilde{\boldsymbol{F}}^{\mathrm{H}}\boldsymbol{a}_{\mathrm{BS}}(\phi_{\ell,1})\right)\right]+\boldsymbol{\mathcal{Z}}_1, \tag{15}$$

where $\boldsymbol{u}(\tau_{\ell,1})=[e^{-\mathrm{j}2\pi\gamma\tau_{\ell,1}\frac{1}{f_{\mathrm{s}}}},e^{-\mathrm{j}2\pi\gamma\tau_{\ell,1}\frac{2}{f_{\mathrm{s}}}},\cdots,e^{-\mathrm{j}2\pi\gamma\tau_{\ell,1}\frac{K}{f_{\mathrm{s}}}}]^{\mathrm{T}}$ is the fast-time sinusoidal vector, $\boldsymbol{v}(f_{\ell,1})=[e^{\mathrm{j}2\pi T_{\mathrm{r}}f_{\ell,1}},e^{\mathrm{j}2\pi 2T_{\mathrm{r}}f_{\ell,1}},\cdots,e^{\mathrm{j}2\pi MT_{\mathrm{r}}f_{\ell,1}}]^{\mathrm{T}}$ is the slow-time sinusoidal vector corresponding to different chirps at the channel estimation stage, $\tilde{\boldsymbol{F}}=[\tilde{\boldsymbol{f}}_{\mathrm{RF},1},\tilde{\boldsymbol{f}}_{\mathrm{RF},2},\cdots,\tilde{\boldsymbol{f}}_{\mathrm{RF},M}]$ contains beamforming vectors for all the FMCW chirps, and $\boldsymbol{\mathcal{Z}}_1$ denotes the noise tensor. CP decomposition can then be performed on $\boldsymbol{\mathcal{Y}}_1$ to estimate the parameters $\{\hat{\beta}_\ell,\hat{\theta}_{\ell,1},\hat{\phi}_{\ell,1},\hat{\tau}_{\ell,1},\hat{f}_{\ell,1}\}$. It is illustrated in Fig. 4. In particular, let $\boldsymbol{b}_1(\ell)=\tilde{\boldsymbol{a}}_{\mathrm{MS}}(\theta_{\ell,1})$, $\boldsymbol{b}_2(\ell)=\tilde{\beta}_\ell\boldsymbol{u}(\tau_{\ell,1})$, and $\boldsymbol{b}_3(\ell)=\boldsymbol{v}(f_{\ell,1})\odot(\tilde{\boldsymbol{F}}^{\mathrm{H}}\boldsymbol{a}_{\mathrm{BS}}(\phi_{\ell,1}))$ denote the mode-1, mode-2, and mode-3 fibers of the rank-1 components, respectively. By stacking them across all L paths, we form the factor matrices $\boldsymbol{B}_1=[\boldsymbol{b}_1(1),\boldsymbol{b}_1(2),\cdots,\boldsymbol{b}_1(L)]$, $\boldsymbol{B}_2=[\boldsymbol{b}_2(1),\boldsymbol{b}_2(2),\cdots,\boldsymbol{b}_2(L)]$, and $\boldsymbol{B}_3=[\boldsymbol{b}_3(1),\boldsymbol{b}_3(2),\cdots,\boldsymbol{b}_3(L)]$.

To exploit the structural prior knowledge and improve robustness, we impose a Vandermonde structure on the second factor matrix $\boldsymbol{B}_2$. This structure arises from the shift-invariance property of $\boldsymbol{b}_2(\ell)$ due to the uniform subcarrier spacing. As such, the CP decomposition of $\boldsymbol{\mathcal{Y}}_1$ can be formulated as the following least squares optimization problem

$$\min_{\hat{\boldsymbol{B}}_1,\hat{\boldsymbol{B}}_2\in\mathcal{V},\hat{\boldsymbol{B}}_3}\left\|\boldsymbol{\mathcal{Y}}_1-[\![\hat{\boldsymbol{B}}_1,\hat{\boldsymbol{B}}_2,\hat{\boldsymbol{B}}_3]\!]\right\|_{\mathrm{F}}^2, \tag{16}$$

where $\mathcal{V}$ denotes the set of Vandermonde matrices, while $\hat{\boldsymbol{B}}_1=[\hat{\boldsymbol{b}}_1(1),\hat{\boldsymbol{b}}_1(2),\cdots,\hat{\boldsymbol{b}}_1(L)]$, $\hat{\boldsymbol{B}}_2=[\hat{\boldsymbol{b}}_2(1),\hat{\boldsymbol{b}}_2(2),\cdots,\hat{\boldsymbol{b}}_2(L)]$, and $\hat{\boldsymbol{B}}_3=[\hat{\boldsymbol{b}}_3(1),\hat{\boldsymbol{b}}_3(2),\cdots,\hat{\boldsymbol{b}}_3(L)]$ are the estimated factor matrices. The above CP decomposition problem can be solved by the alternating least squares (ALS) algorithm [46]. Since the multiple parameters are automatically associated with each scatterer point in the decomposed tensor factors [47], [48], a post-processing step to match these parameters can be avoided.

B. Determination of Scatterer Locations and Velocities

To determine the coordinate of a scatterer point, its corresponding direction and distance are required. Specifically, the AOA and propagation delay are estimated through solving the following maximization problems

$$\hat{\theta}_{\ell,1}=\arg\max_{\theta}\frac{\left|\hat{\boldsymbol{b}}_1^{\mathrm{H}}(\ell)\tilde{\boldsymbol{a}}_{\mathrm{MS}}(\theta)\right|^2}{\left\|\hat{\boldsymbol{b}}_1^{\mathrm{H}}(\ell)\right\|_2^2\|\tilde{\boldsymbol{a}}_{\mathrm{MS}}(\theta)\|_2^2} \tag{17}$$

and

$$\hat{\tau}_{\ell,1}=\arg\max_{\tau}\frac{\left|\hat{\boldsymbol{b}}_2^{\mathrm{H}}(\ell)\boldsymbol{u}(\tau)\right|^2}{\left\|\hat{\boldsymbol{b}}_2^{\mathrm{H}}(\ell)\right\|_2^2\|\boldsymbol{u}(\tau)\|_2^2}, \tag{18}$$

respectively. The solutions of (17) and (18) can be found by one-dimensional searches over θ and τ, respectively. Then, the estimated length of the ℓ-th propagation path can be calculated as $\hat{d}_{\ell,1}=c\hat{\tau}_{\ell,1}$.

While the conventional methods tailored for stationary scenarios ignore Doppler and use a one-dimensional search for AOD estimation [37], the high-dynamic scenarios considered here exhibit coupling between AOD and velocity in $\hat{\boldsymbol{B}}_3$, thereby complicating their joint estimation. Existing iterative methods can decouple AOD and velocity but still suffer from high computational complexity due to high complexity per iteration [49]. In this regard, we propose to exploit the geometry relationship between the MS, BS, and the scatterer point as shown in Fig. 3, yielding the closed-form solution of AOD estimation as

$$\hat{\phi}_{\ell,1}=\arccos\left(\frac{\hat{d}_{\ell,1}^2+\hat{g}_{\ell,1}^2-d_{\mathrm{bis}}^2}{2\hat{d}_{\ell,1}\hat{g}_{\ell,1}}\right)$$
$$-\arccos\left(\frac{d_{\mathrm{bis}}^2+\hat{g}_{\ell,1}^2-\hat{d}_{\ell,1}^2}{2d_{\mathrm{bis}}\hat{g}_{\ell,1}}\right). \tag{19}$$

Here,

$$\hat{g}_{\ell,1}=\sqrt{\hat{d}_{\ell,1}^2+d_{\mathrm{bis}}^2-2\hat{d}_{\ell,1}d_{\mathrm{bis}}\cos\hat{\theta}_{\ell,1}} \tag{20}$$

denotes the distance between the BS and the auxiliary point D in Fig. 3 located at a distance $\hat{d}_{\ell,1}$ in the direction of AOA $\hat{\theta}_{\ell,1}$, and d_{bis} denotes the distance between the MS and BS, which can be obtained from the time-of-flight measurement or Global Positioning System. The localization $[\hat{x}_{\ell,1},\hat{y}_{\ell,1}]^{\mathrm{T}}$ of the ℓ-th scatterer can then be represented as

$$\begin{bmatrix}\hat{x}_{\ell,1}\\ \hat{y}_{\ell,1}\end{bmatrix}=\frac{\hat{d}_{\ell,1}\sin(|\hat{\theta}_{p,1}|)}{\sin(|\hat{\theta}_{\ell,1}|)+\sin(|\hat{\phi}_{\ell,1}|)}\begin{bmatrix}\sin(\hat{\phi}_{\ell,1})\\ \cos(\hat{\phi}_{\ell,1})\end{bmatrix}. \tag{21}$$

Subsequently, the Doppler frequency of the path corresponding to the ℓ-th scatterer point can be estimated as

$$\hat{f}_{\ell,1}=\arg\max_{f}\frac{\left|\hat{\boldsymbol{b}}_3^{\mathrm{H}}(\ell)\left[\boldsymbol{v}(f)\odot\left(\tilde{\boldsymbol{F}}^{\mathrm{H}}\boldsymbol{a}_{\mathrm{BS}}(\hat{\phi}_{\ell,1})\right)\right]\right|^2}{\left\|\hat{\boldsymbol{b}}_3^{\mathrm{H}}(\ell)\right\|_2^2\left\|\boldsymbol{v}(f)\odot\left(\tilde{\boldsymbol{F}}^{\mathrm{H}}\boldsymbol{a}_{\mathrm{BS}}(\hat{\phi}_{\ell,1})\right)\right\|_2^2}. \tag{22}$$

Substituting (12) into (22), the bistatic velocity of the ℓ-th scatterer point is calculated as

$$\hat{v}_\ell=\frac{\lambda\hat{f}_{\ell,1}}{2\sin((|\hat{\theta}_{\ell,1}|+|\hat{\phi}_{\ell,1}|)/2)}. \tag{23}$$

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

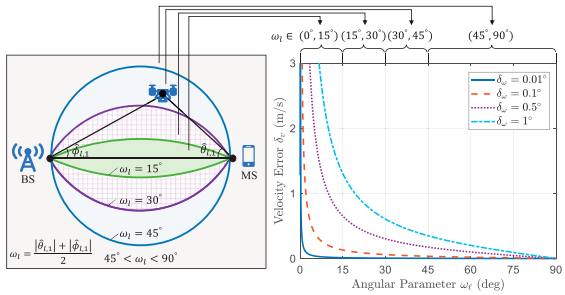


Fig. 5. Illustration of relationship between the velocity estimation error δ_v and the angular estimation error δ_ω corresponding to various geometrical locations for a target moving at the velocity of 20 m/s.

Here, the accuracy of $\hat{v}_\ell$ is related to the accuracy of $\hat{\theta}_{\ell,1}$ and $\hat{\phi}_{\ell,1}$. To quantify how angular uncertainty propagates to the velocity estimate, we employ first-order error analysis. Defining $\omega_\ell = (|\hat{\theta}_{\ell,1}| + |\hat{\phi}_{\ell,1}|)/2$, the relationship between the velocity error δ_v of v_ℓ and the angular error δ_ω of ω_ℓ can be represented by

$$\delta_v \approx \left|\frac{d\hat{v}_\ell}{d\omega_\ell}\right| \delta_\omega = \frac{\lambda \hat{f}_{\ell,1}}{2} \left|\frac{d}{d\omega_\ell}\left(\frac{1}{\sin\omega_\ell}\right)\right| \delta_\omega = \delta_\omega\, \hat{v}_\ell \cot\omega_\ell. \tag{24}$$

As shown in Fig. 5, the sensitivity to angular errors is governed by the factor $\cot\omega_\ell$, which decreases as ω_ℓ increases. Its practical impact is negligible due to the small spatial extent of such conditions. Moreover, these conditions can be identified via angle estimations. Thus, velocity estimation remains robust under typical angular estimation errors.

Based on the estimates $\{\hat{\theta}_{\ell,1}, \hat{\phi}_{\ell,1}, \hat{\tau}_{\ell,1}, \hat{f}_{\ell,1}\}$, the complex path gain for the ℓ-th path is calculated by

$$\begin{aligned}
\hat{\beta}_\ell &= [\boldsymbol{\Gamma}_1]_{(\ell,\ell)}[\boldsymbol{\Gamma}_2]_{(\ell,\ell)}[\boldsymbol{\Gamma}_3]_{(\ell,\ell)} e^{\mathrm{j}2\pi f_c \hat{\tau}_{\ell,1}},\\
\boldsymbol{\Gamma}_1 &= [\hat{\boldsymbol{B}}_1]^{\mathrm{H}}_{(\cdot,\ell)} \tilde{\boldsymbol{a}}_{\mathrm{MS}}(\hat{\theta}_{\ell,1}),\\
\boldsymbol{\Gamma}_2 &= [\hat{\boldsymbol{B}}_2]^{\mathrm{H}}_{(\cdot,\ell)} \boldsymbol{u}\left(\hat{\tau}_{\ell,1}\right),\\
\boldsymbol{\Gamma}_3 &= [\hat{\boldsymbol{B}}_3]^{\mathrm{H}}_{(\cdot,\ell)} \left(\boldsymbol{v}(\hat{f}_{\ell,1}) \odot (\tilde{\boldsymbol{F}}^{\mathrm{H}} \boldsymbol{a}_{\mathrm{BS}}(\hat{\phi}_{\ell,1}))\right).
\end{aligned} \tag{25}$$

C. Analysis of Mobility-Induced Location Deviation

Since the angle δ_ℓ between the bistatic velocity v_ℓ and the actual velocity $\bar{v}_\ell$ in (9) is unknown in bistatic processing [50], the full motion status of the scatterers cannot be obtained, preventing an accurate estimation of $\theta_{\ell,g}$, $\phi_{\ell,g}$, and $d_{\ell,g}$ in (10). Hence, the channel matrix $\hat{\boldsymbol{H}}_{k,g}(m)$ is estimated based on only the feedback parameters $\hat{\theta}_{\ell,1}$, $\hat{\phi}_{\ell,1}$, and $\hat{d}_{\ell,1}$ as

$$\begin{aligned}
\hat{\boldsymbol{H}}_{k,g}(m) = &\sum_{\ell=1}^{L} \hat{\beta}_\ell e^{\mathrm{j}2\pi(mT_{\mathrm{s}}+M(g-1)T_{\mathrm{s}}+MT_{\mathrm{r}})\hat{f}_{\ell,1}}\\
&\times e^{-\mathrm{j}2\pi k\hat{\tau}_{\ell,1}\Delta} \boldsymbol{a}_{\mathrm{MS}}\left(\hat{\theta}_{\ell,1}\right) \boldsymbol{a}^{\mathrm{H}}_{\mathrm{BS}}\left(\hat{\phi}_{\ell,1}\right).
\end{aligned} \tag{26}$$

However, the location parameters estimated in the first subframe differ from the actual parameters in the g-th subframe, leading to notable discrepancy between the estimated channel $\hat{\boldsymbol{H}}_{k,g}(m)$ and the actual channel $\boldsymbol{H}_{k,g}(m)$ in (13). Specifically, the differences between the actual parameters in the g-th subframe and their estimations can be characterized by

$$\begin{aligned}
\theta_{\ell,g} - \hat{\theta}_{\ell,1} &= (\theta_{\ell,1} + \tilde{\theta}_{\ell,g}) - (\theta_{\ell,1} - \Delta\theta_{\ell,1}) = \tilde{\theta}_{\ell,g} + \Delta\theta_{\ell,1},\\
\phi_{\ell,g} - \hat{\phi}_{\ell,1} &= (\phi_{\ell,1} + \tilde{\phi}_{\ell,g}) - (\phi_{\ell,1} - \Delta\phi_{\ell,1}) = \tilde{\phi}_{\ell,g} + \Delta\phi_{\ell,1},\\
d_{\ell,g} - \hat{d}_{\ell,1} &= (d_{\ell,1} + \tilde{d}_{\ell,g}) - (d_{\ell,1} - \Delta d_{\ell,1}) = \tilde{d}_{\ell,g} + \Delta d_{\ell,1},
\end{aligned} \tag{27}$$

respectively. Here, $\Delta\theta_{\ell,1} = \theta_{\ell,1} - \hat{\theta}_{\ell,1}$, $\Delta\phi_{\ell,1} = \phi_{\ell,1} - \hat{\phi}_{\ell,1}$, and $\Delta d_{\ell,1} = d_{\ell,1} - \hat{d}_{\ell,1}$ denote the estimation error in the first subframe, and $\tilde{\theta}_{\ell,g} = \theta_{\ell,g} - \theta_{\ell,1}$, $\tilde{\phi}_{\ell,g} = \phi_{\ell,g} - \phi_{\ell,1}$, $\tilde{d}_{\ell,g} = d_{\ell,g} - d_{\ell,1}$ denote the parameter deviation from the first subframe to the g-th subframe, respectively. As such, the total estimation errors can be significant for the parameters associated with high-velocity moving targets, even when the estimation error $\Delta\theta_{\ell,1}$, $\Delta\phi_{\ell,1}$, and $\Delta d_{\ell,1}$ is sufficiently small.

For subsequent precoding design, we distinguish mobile targets from background scatterers. This is achieved by identifying the P scatterers with the highest estimated velocity among the L total as moving targets, while the remaining $L-P$ are classified as background scatterers. Since background scatterers exhibit less location deviation, reliable data transmission can be guaranteed by focusing the transmission energy on the paths corresponding to these quasi-static scatterers.

IV. Sensing-Aided Precoding Design

In this section, we first discuss the performance degradation of conventional precoding designs in high-dynamic MIMO channels. As a remedy, we propose a sensing-aided precoding design based on a channel division criterion, where the moving target identification is leveraged to select the propagation paths that support reliable data transmission.

A. Conventional Precoding Design

We analyze the achievable rate of the conventional precoding methods to evaluate their transmission quality. When Gaussian symbols are transmitted over the time-varying channel $\boldsymbol{H}_{k,g}(m)$ in (13), the achievable rate within period of the m-th transmitted symbol in the g-th subframe is expressed as

$$\mathcal{R}_g(m) = \frac{1}{K}\sum_{k=1}^{K} \log_2\left(\left|\boldsymbol{I}_{N_{\mathrm{s}}} + \frac{\varrho}{N_{\mathrm{s}}} \boldsymbol{R}_{\mathrm{n}}^{-1}(m) \bar{\boldsymbol{H}}_{k,g}(m) \bar{\boldsymbol{H}}^{\mathrm{H}}_{k,g}(m)\right|\right), \tag{28}$$

where

$$\bar{\boldsymbol{H}}_{k,g}(m) = \boldsymbol{W}^{\mathrm{H}}_{k,g}(m) \boldsymbol{H}_{k,g}(m) \boldsymbol{F}_{k,g}(m) \tag{29}$$

is defined as the equivalent channel matrix after precoding and combining. In (29), $\boldsymbol{F}_{k,g}(m) = \boldsymbol{F}_{\mathrm{RF}} \boldsymbol{F}_{\mathrm{BB},k,g}(m)$ and $\boldsymbol{W}_{k,g}(m) = \boldsymbol{W}_{\mathrm{RF}} \boldsymbol{W}_{\mathrm{BB},k,g}(m)$ denote respectively the hybrid precoding and combining matrices, ϱ denotes the average received power, and $\boldsymbol{R}_{\mathrm{n}}(m) = \sigma_{\mathrm{n}}^2 \boldsymbol{W}^{\mathrm{H}}_{k,g}(m) \boldsymbol{W}_{k,g}(m)$ denotes

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

the noise covariance matrix. Theoretically, the precoding matrices $\boldsymbol{F}_{\mathrm{RF}}$, $\boldsymbol{F}_{\mathrm{BB},k,g}(m)$ and the combining matrices $\boldsymbol{W}_{\mathrm{RF}}$, $\boldsymbol{W}_{\mathrm{BB},k,g}(m)$ in both analog and digital domains can be jointly designed to maximize the achievable rate as

$$\begin{aligned}\max_{\boldsymbol{F}_{\mathrm{RF}},\boldsymbol{F}_{\mathrm{BB},k,g},\boldsymbol{W}_{\mathrm{RF}},\boldsymbol{W}_{\mathrm{BB},k,g}} \quad & \mathcal{R}_g(m)\\ \text{s.t.} \quad & \left|[\boldsymbol{F}_{\mathrm{RF}}]_{(i,j)}\right| = \frac{1}{\sqrt{N_{\mathrm{BS}}}}, \forall i,j,\\ & \left|[\boldsymbol{W}_{\mathrm{RF}}]_{(i,j)}\right| = \frac{1}{\sqrt{N_{\mathrm{MS}}}}, \forall i,j,\\ & ||\boldsymbol{F}_{\mathrm{RF}}\boldsymbol{F}_{\mathrm{BB},k,g}(m)||_{\mathrm{F}}^2 = N_{\mathrm{s}},\\ & ||\boldsymbol{W}_{\mathrm{RF}}\boldsymbol{W}_{\mathrm{BB},k,g}(m)||_{\mathrm{F}}^2 = N_{\mathrm{s}}. \end{aligned} \tag{30}$$

Since the optimization problem in (30) is non-convex, the sub-optimal solutions are considered. Typically, the hybrid precoding and combining matrices are obtained by minimizing the Frobenius norm between the optimal unconstrained solutions and the hybrid solutions [51], i.e.,

$$\begin{aligned}(\boldsymbol{F}_{\mathrm{RF}}, \boldsymbol{F}_{\mathrm{BB},k,g}) &= \underset{\boldsymbol{F}_{\mathrm{RF}},\boldsymbol{F}_{\mathrm{BB},k,g}}{\arg\min} \|\boldsymbol{F}_{\mathrm{opt},k,g}(m)\\ &\qquad -\boldsymbol{F}_{\mathrm{RF}}\boldsymbol{F}_{\mathrm{BB},k,g}\|_{\mathrm{F}},\\ (\boldsymbol{W}_{\mathrm{RF}}, \boldsymbol{W}_{\mathrm{BB},k,g}) &= \underset{\boldsymbol{W}_{\mathrm{RF}},\boldsymbol{W}_{\mathrm{BB},k,g}}{\arg\min} \|\boldsymbol{W}_{\mathrm{opt},k,g}(m)\\ &\qquad -\boldsymbol{W}_{\mathrm{RF}}\boldsymbol{W}_{\mathrm{BB},k,g}\|_{\mathrm{F}}, \end{aligned} \tag{31}$$

where

$$\begin{aligned}\boldsymbol{F}_{\mathrm{opt},k,g}(m) &= [\boldsymbol{V}_{k,g}]_{(\cdot,1:N_{\mathrm{s}})},\\ \boldsymbol{W}_{\mathrm{opt},k,g}(m) &= [\boldsymbol{U}_{k,g}]_{(\cdot,1:N_{\mathrm{s}})} \end{aligned} \tag{32}$$

are the optimal unconstrained precoding and combining matrices calculated from the compact singular value decomposition (SVD) of the estimated channel matrix $\hat{\boldsymbol{H}}_{k,g}(m)$ in (26) as

$$\hat{\boldsymbol{H}}_{k,g}(m) = \boldsymbol{U}_{k,g}\boldsymbol{\Sigma}_{k,g}\boldsymbol{V}_{k,g}^{\mathrm{H}}. \tag{33}$$

Here, $\boldsymbol{U}_{k,g} \in \mathbb{C}^{N_{\mathrm{MS}}\times L}$ and $\boldsymbol{V}_{k,g} \in \mathbb{C}^{N_{\mathrm{BS}}\times L}$ are semi-unitary matrices, and $\boldsymbol{\Sigma}_{k,g}$ is a square diagonal matrix.[3]

In particular, the achievable rate in (28) corresponding to the optimal unconstrained design in (32) can be calculated as

$$\begin{aligned}&\mathcal{R}_{\mathrm{opt},g}(m) =\\ &\frac{1}{K}\sum_{k=1}^{K}\log_2\left(\left|\boldsymbol{I}_{N_{\mathrm{s}}} + \frac{\varrho}{N_{\mathrm{s}}\sigma_{\mathrm{n}}^2}\left(\boldsymbol{W}_{\mathrm{opt},k,g}^{\mathrm{H}}\boldsymbol{W}_{\mathrm{opt},k,g}\right)^{-1}\right.\right.\\ &\qquad\left.\left.\times\bar{\boldsymbol{H}}_{\mathrm{opt},k,g}(m)\bar{\boldsymbol{H}}_{\mathrm{opt},k,g}^{\mathrm{H}}(m)\right|\right), \end{aligned} \tag{34}$$

where $\bar{\boldsymbol{H}}_{\mathrm{opt},k,g}(m) = \boldsymbol{W}_{\mathrm{opt},k,g}^{\mathrm{H}}(m)\boldsymbol{H}_{k,g}(m)\boldsymbol{F}_{\mathrm{opt},k,g}(m)$ denotes the corresponding equivalent channel matrix. When the scatterer points remain stationary across subsequent subframes, the estimated channel in (33) approaches the actual one, i.e., $\hat{\boldsymbol{H}}_{k,g}(m) = \boldsymbol{H}_{k,g}(m)$. In that case, we have $\bar{\boldsymbol{H}}_{\mathrm{opt},k,g}(m) = [\boldsymbol{\Sigma}_{k,g}]_{(1:N_{\mathrm{s}},1:N_{\mathrm{s}})}$ by substituting (32) into (29). Moreover, each eigenvalue in $[\boldsymbol{\Sigma}_{k,g}]_{(1:N_{\mathrm{s}},1:N_{\mathrm{s}})}$ corresponds to a sub-channel supporting an independent data stream. As such, the highest instantaneous achievable rate corresponding to the optimal design becomes $\mathcal{R}_{\mathrm{opt},g}(m) = \frac{1}{K}\sum_{k=1}^{K}\log_2\left(\left|\boldsymbol{I}_{N_{\mathrm{s}}} + \frac{\varrho}{N_{\mathrm{s}}\sigma_{\mathrm{n}}^2}[\boldsymbol{\Sigma}_{k,g}]_{(1:N_{\mathrm{s}},1:N_{\mathrm{s}})}^{\mathrm{H}}[\boldsymbol{\Sigma}_{k,g}]_{(1:N_{\mathrm{s}},1:N_{\mathrm{s}})}\right|\right)$.

However, in high-dynamic scenarios, the approximation from $\hat{\boldsymbol{H}}_{k,g}(m)$ to $\boldsymbol{H}_{k,g}(m)$ is difficult even when SNR is high. As demonstrated in (26), $\hat{\boldsymbol{H}}_{k,g}(m)$ is composed of parameters estimated in the first subframe, while these parameters continuously change across subframes especially for those of the high-velocity moving targets. Severe channel fluctuations can be induced by such parameter deviation, and the equivalent channel matrix $\bar{\boldsymbol{H}}_{\mathrm{opt},k,g}(m)$ corresponding to the optimal design is not diagonal anymore, resulting in the inter-channel interferences and degraded transmission performance.

B. Proposed Precoding Design

To mitigate the negative impact of the dynamic propagation paths, the energy can be concentrated on the propagation paths corresponding to quasi-static background scatterers. To this end, a channel division criterion is designed to partition the channel matrix into two parts characterizing mobile and quasi-static propagation paths, based on which the property of digital-analog dual orthogonality is investigated to facilitate an effective precoding design.

1) Analog Precoding and Combining Matrices: By exploiting the sensing results of moving target identification in Section III, the estimated channel matrix $\hat{\boldsymbol{H}}_{k,g}(m)$ in (26) can be reformulated as

$$\begin{aligned}\hat{\boldsymbol{H}}_{k,g}(m) &= \boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})\hat{\boldsymbol{\Xi}}_{k,g}(m)\boldsymbol{A}_{\mathrm{BS}}^{\mathrm{H}}(\hat{\boldsymbol{\phi}})\\ &= [\boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})]_{(\cdot,1:P)}[\hat{\boldsymbol{\Xi}}_{k,g}]_{(1:P,1:P)}[\boldsymbol{A}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}})]_{(\cdot,1:P)}^{\mathrm{H}}\\ &\quad + [\boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})]_{(\cdot,P+1:L)}[\hat{\boldsymbol{\Xi}}_{k,g}]_{(P+1:L,P+1:L)}[\boldsymbol{A}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}})]_{(\cdot,P+1:L)}^{\mathrm{H}}, \end{aligned} \tag{35}$$

where $\boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}}) = [\boldsymbol{a}_{\mathrm{MS}}(\hat{\theta}_{1,1}), \boldsymbol{a}_{\mathrm{MS}}(\hat{\theta}_{2,1}), \cdots, \boldsymbol{a}_{\mathrm{MS}}(\hat{\theta}_{L,1})]$ and $\boldsymbol{A}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}}) = [\boldsymbol{a}_{\mathrm{BS}}(\hat{\phi}_{1,1}), \boldsymbol{a}_{\mathrm{BS}}(\hat{\phi}_{2,1}), \cdots, \boldsymbol{a}_{\mathrm{BS}}(\hat{\phi}_{L,1})]$ denote respectively the estimations of receive and transmit steering matrices in the first subframe,[4] and

$$\hat{\boldsymbol{\Xi}}_{k,g}(m) = \mathrm{diag}(\hat{\zeta}_{1,k,g}, \hat{\zeta}_{2,k,g}, \cdots, \hat{\zeta}_{L,k,g}) \in \mathbb{C}^{L\times L} \tag{36}$$

is defined as the *effective channel* with its diagonal element being

$$\hat{\zeta}_{\ell,k,g} = \hat{\beta}_{\ell}e^{\mathrm{j}2\pi(mT_{\mathrm{s}}+(g-1)MT_{\mathrm{s}}+MT_{\mathrm{r}})\hat{f}_{\ell,1}}e^{-\mathrm{j}2\pi k\hat{\tau}_{\ell,1}\Delta}. \tag{37}$$

As such, the mobile and quasi-static propagation paths are decoupled in the angular domain, since the steering vectors

[3]Since the hybrid precoding and combining matrices can be formulated according to the same principle when the CSI is available at both the BS and the MS, the precoding and combining is referred to as the precoding design for brevity in this paper. However, we still present the derivation of combining matrices to demonstrate the time-varying achievable rate corresponding to different designs.

[4]For notational simplicity, the estimated and true indices are assumed to be the same for each scatterer point, while the subsequent precoding design does not rely on the correspondence between the estimated and true indices.

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

corresponding to different AOAs and AODs are asymptotically orthogonal when the number of antennas tends to infinity [51].

Accordingly, the steering matrices corresponding to the mobile and quasi-static propagation paths are orthogonal when the number of antennas is large enough, i.e.,

$$[\boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})]^{\mathrm{H}}_{(\cdot,1:P)}[\boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})]_{(\cdot,P+1:L)} \to \mathbf{0}, N_{\mathrm{MS}} \to \infty,$$
$$[\boldsymbol{A}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}})]^{\mathrm{H}}_{(\cdot,1:P)}[\boldsymbol{A}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}})]_{(\cdot,P+1:L)} \to \mathbf{0}, N_{\mathrm{BS}} \to \infty. \quad (38)$$

Thus, the analog precoding and combining matrices can be respectively set as

$$\boldsymbol{F}_{\mathrm{RF}} = \left[[\boldsymbol{A}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}})]_{(\cdot,P+1:P+N_{\mathrm{s}})}, \mathbf{0}_{N_{\mathrm{BS}}\times(M_{\mathrm{BS}}-N_{\mathrm{s}})}\right],$$
$$\boldsymbol{W}_{\mathrm{RF}} = \left[[\boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})]_{(\cdot,P+1:P+N_{\mathrm{s}})}, \mathbf{0}_{N_{\mathrm{MS}}\times(M_{\mathrm{MS}}-N_{\mathrm{s}})}\right] \quad (39)$$

to eliminate the data transmission towards the directions of mobile propagation paths.

2) Digital Precoding and Combining Matrices: When the numbers of transmit and receive antennas are relatively small, the energy leakage into mobile propagation paths is further suppressed by the digital-domain processing, thereby reducing the influence of propagation paths associated with moving targets. To be specific, the effective channel $\hat{\boldsymbol{\Xi}}_{k,g}(m)$ is constructed with the estimated parameters $\{\hat{\beta}_\ell, \hat{f}_{\ell,1}, \hat{\tau}_{\ell,1}\}$ as shown in (36) and (37), and an orthogonal basis of $\hat{\boldsymbol{\Xi}}_{k,g}(m)$ is obtained through its SVD as

$$\hat{\boldsymbol{\Xi}}_{k,g}(m) = \tilde{\boldsymbol{U}}_{k,g}(m)\tilde{\boldsymbol{\Psi}}_{k,g}(m)\tilde{\boldsymbol{V}}^{\mathrm{H}}_{k,g}(m), \quad (40)$$

where $\tilde{\boldsymbol{U}}_{k,g}(m)$ and $\tilde{\boldsymbol{V}}_{k,g}(m)$ are unitary matrices, and $\tilde{\boldsymbol{\Psi}}_{k,g}(m)$ is a real square diagonal matrix. To keep the notation simple, we will drop the symbol index m in $\hat{\boldsymbol{\Xi}}_{k,g}(m)$, $\tilde{\boldsymbol{U}}_{k,g}(m)$, $\tilde{\boldsymbol{\Psi}}_{k,g}(m)$ and $\tilde{\boldsymbol{V}}_{k,g}(m)$ hereafter. Thanks to the diagonal property of $\hat{\boldsymbol{\Xi}}_{k,g}$, each of its singular vectors and singular values uniquely corresponds to an individual propagation path. As such, the parameters associated with mobile and quasi-static propagation paths remain separable in the digital domain. However, the descending ordering of the singular values in $\tilde{\boldsymbol{\Psi}}_{k,g}$ hinders the direct identification of mobile paths, as the indices of these paths are no longer sequentially aligned from 1 to P. To retain the alignment between corresponding columns in $\boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})$, $\boldsymbol{A}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}})$, and $\tilde{\boldsymbol{\Psi}}_{k,g}$, the following characteristics of $\tilde{\boldsymbol{V}}_{k,g}$ can be exploited.

Theorem 1: The $\tilde{\boldsymbol{V}}_{k,g}$ is either a diagonal matrix with ± 1 on its main diagonal or a transformed version of such diagonal matrix through a sequence of elementary transformations.

Proof: See Appendix A. ■

According to the characteristics of $\tilde{\boldsymbol{V}}_{k,g}$ in *Theorem 1*, the position exchanges of the singular values in $\tilde{\boldsymbol{\Psi}}_{k,g}$ are preserved in $\tilde{\boldsymbol{V}}_{k,g}$, and thus, two index vectors, i.e.,

$$\boldsymbol{\nu}_{(p)} = \arg\max_{\boldsymbol{\nu}_{(p)}} \left(\left|[\tilde{\boldsymbol{V}}_{k,g}]_{(p,\boldsymbol{\nu}_{(p)})}\right|\right), \forall p = 1, 2, \cdots, P,$$
$$\acute{\boldsymbol{\nu}}_{(q)} = \arg\max_{\acute{\boldsymbol{\nu}}_{(q)}} \left(\left|[\tilde{\boldsymbol{V}}_{k,g}]_{(q,\acute{\boldsymbol{\nu}}_{(q)})}\right|\right), \forall q = P+1, P+2, \cdots, L, \quad (41)$$

can be defined to identify the indices of moving targets and background scatterers in $\tilde{\boldsymbol{\Psi}}_{k,g}$, respectively. Incorporating the definitions of $\boldsymbol{\nu}$ and $\acute{\boldsymbol{\nu}}$, the estimated channel matrix $\hat{\boldsymbol{H}}_{k,g}(m)$ in (26) can be reformulated from (35) as

$$\begin{aligned}\hat{\boldsymbol{H}}_{k,g}(m) &= \boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})\tilde{\boldsymbol{U}}_{k,g}\tilde{\boldsymbol{\Psi}}_{k,g}\tilde{\boldsymbol{V}}^{\mathrm{H}}_{k,g}\boldsymbol{A}^{\mathrm{H}}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}}) \\ &= \boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})[\tilde{\boldsymbol{U}}_{k,g}]_{(\cdot,\boldsymbol{\nu})}[\tilde{\boldsymbol{\Psi}}_{k,g}]_{(\boldsymbol{\nu},\boldsymbol{\nu})}[\tilde{\boldsymbol{V}}_{k,g}]^{\mathrm{H}}_{(\cdot,\boldsymbol{\nu})}\boldsymbol{A}^{\mathrm{H}}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}}) \\ &\quad + \boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})[\tilde{\boldsymbol{U}}_{k,g}]_{(\cdot,\acute{\boldsymbol{\nu}})}[\tilde{\boldsymbol{\Psi}}_{k,g}]_{(\acute{\boldsymbol{\nu}},\acute{\boldsymbol{\nu}})}[\tilde{\boldsymbol{V}}_{k,g}]^{\mathrm{H}}_{(\cdot,\acute{\boldsymbol{\nu}})}\boldsymbol{A}^{\mathrm{H}}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}}).\end{aligned} \quad (42)$$

Theorem 2: The divisions of the estimated channel matrix $\hat{\boldsymbol{H}}_{k,g}(m)$ represented in (42) and (35) are equivalent, i.e.,

$$\begin{aligned}&\boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})[\tilde{\boldsymbol{U}}_{k,g}]_{(\cdot,\boldsymbol{\nu})}[\tilde{\boldsymbol{\Psi}}_{k,g}]_{(\boldsymbol{\nu},\boldsymbol{\nu})}[\tilde{\boldsymbol{V}}_{k,g}]^{\mathrm{H}}_{(\cdot,\boldsymbol{\nu})}\boldsymbol{A}^{\mathrm{H}}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}}) \\ &\quad = [\boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})]_{(\cdot,1:P)}[\hat{\boldsymbol{\Xi}}_{k,g}]_{(1:P,1:P)}[\boldsymbol{A}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}})]^{\mathrm{H}}_{(\cdot,1:P)}\end{aligned} \quad (43)$$

and

$$\begin{aligned}&\boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})[\tilde{\boldsymbol{U}}_{k,g}]_{(\cdot,\acute{\boldsymbol{\nu}})}[\tilde{\boldsymbol{\Psi}}_{k,g}]_{(\acute{\boldsymbol{\nu}},\acute{\boldsymbol{\nu}})}[\tilde{\boldsymbol{V}}_{k,g}]^{\mathrm{H}}_{(\cdot,\acute{\boldsymbol{\nu}})}\boldsymbol{A}^{\mathrm{H}}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}}) \\ &= [\boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})]_{(\cdot,P+1:L)}[\hat{\boldsymbol{\Xi}}_{k,g}]_{(P+1:L,P+1:L)}[\boldsymbol{A}_{\mathrm{BS}}(\hat{\boldsymbol{\phi}})]^{\mathrm{H}}_{(\cdot,P+1:L)}.\end{aligned} \quad (44)$$

Proof: See Appendix B. ■

Based on *Theorem 2*, the data transmission leakage over mobile propagation paths can be further eliminated by leveraging the orthogonality between different columns in the unitary matrices $\tilde{\boldsymbol{U}}_{k,g}$ and $\tilde{\boldsymbol{V}}_{k,g}$. In particular, the digital precoding and combining matrices can be configured as

$$\begin{aligned}&\boldsymbol{F}_{\mathrm{BB},k,g}(m) \\ &= \begin{cases}[\tilde{\boldsymbol{V}}_{k,g}]_{(1:M_{\mathrm{BS}},\acute{\boldsymbol{\nu}}_{(1:N_{\mathrm{s}})})}, & \text{if } L \geq M_{\mathrm{BS}}, \\ \left[[\tilde{\boldsymbol{V}}_{k,g}]^{\mathrm{T}}_{(\cdot,\acute{\boldsymbol{\nu}}_{(1:N_{\mathrm{s}})})}, \mathbf{0}^{\mathrm{T}}_{(M_{\mathrm{BS}}-L)\times N_{\mathrm{s}}}\right]^{\mathrm{T}}, & \text{if } L < M_{\mathrm{BS}},\end{cases} \\ &\boldsymbol{W}_{\mathrm{BB},k,g}(m) \\ &= \begin{cases}[\tilde{\boldsymbol{U}}_{k,g}]_{(1:M_{\mathrm{MS}},\acute{\boldsymbol{\nu}}_{(1:N_{\mathrm{s}})})}, & \text{if } L \geq M_{\mathrm{MS}}, \\ \left[[\tilde{\boldsymbol{U}}_{k,g}]^{\mathrm{T}}_{(\cdot,\acute{\boldsymbol{\nu}}_{(1:N_{\mathrm{s}})})}, \mathbf{0}^{\mathrm{T}}_{(M_{\mathrm{MS}}-L)\times N_{\mathrm{s}}}\right]^{\mathrm{T}}, & \text{if } L < M_{\mathrm{MS}},\end{cases}\end{aligned} \quad (45)$$

respectively. Since $\boldsymbol{F}_{\mathrm{BB},k,g}(m)$ and $\boldsymbol{W}_{\mathrm{BB},k,g}(m)$ consist exclusively of components associated with quasi-static scatterers, (45) can be reformulated as

$$\boldsymbol{F}_{\mathrm{BB},k,g}(m) = \left[[\tilde{\boldsymbol{V}}_{k,g}]^{\mathrm{T}}_{(\cdot,\acute{\boldsymbol{\nu}})}, \mathbf{0}^{\mathrm{T}}_{(M_{\mathrm{BS}}-N_{\mathrm{s}})\times N_{\mathrm{s}}}\right]^{\mathrm{T}},$$
$$\boldsymbol{W}_{\mathrm{BB},k,g}(m) = \left[[\tilde{\boldsymbol{U}}_{k,g}]^{\mathrm{T}}_{(\cdot,\acute{\boldsymbol{\nu}})}, \mathbf{0}^{\mathrm{T}}_{(M_{\mathrm{MS}}-N_{\mathrm{s}})\times N_{\mathrm{s}}}\right]^{\mathrm{T}}, \quad (46)$$

when substituting $\hat{\boldsymbol{\Xi}}_{k,g}(m)$ in (40) by

$$\dot{\boldsymbol{\Xi}}_{k,g}(m) = \mathrm{diag}(\hat{\zeta}_{P+1,k,g}, \hat{\zeta}_{P+2,k,g}, \cdots, \hat{\zeta}_{P+N_{\mathrm{s}},k,g}) \quad (47)$$

and following the derivations of index vectors in (41). This substitution leads to a reduction of the feedback at the end of first subframe by enabling the feedback of only a subset of parameters, as discussed in the following remark.

Remark 1: To compute the precoding and combining matrices before the second subframe, it is sufficient to feed back only the parameters associated with the first N_{s} quasi-static scatterers, i.e., $\{\hat{\beta}_{\dot{q}}, \hat{\phi}_{\dot{q},1}, \hat{\theta}_{\dot{q},1}, \hat{f}_{\dot{q},1}, \hat{\tau}_{\dot{q},1}\}$ for $\dot{q} = P+1, P+2, \cdots, P+N_{\mathrm{s}}$. The parameters of the remaining scatterers, which are not required for immediate precoding, can be fed back later with relaxed latency. This reduces the feedback

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Proposed Sensing-aided Precoding Scheme

Input: Received tensor signal $\boldsymbol{\mathcal{Y}}_1$, precoding matrix $\tilde{\boldsymbol{F}}$, combining matrices $\tilde{\boldsymbol{W}}_{\mathrm{BB}}$ and $\tilde{\boldsymbol{W}}_{\mathrm{RF}}$ in the first subframe.

for each $l \in [1, L]$ **do**
  Calculate the estimated AOA $\hat{\theta}_{\ell,1}$ via (17);
  Calculate the estimated time delay $\hat{\tau}_{\ell,1}$ via (18);
  Calculate the estimated AOD $\hat{\phi}_{\ell,1}$ via (19);
  Calculate the location $[\hat{x}_{\ell,1}, \hat{y}_{\ell,1}]^{\mathrm{T}}$ via (21);
  Calculate the estimated velocity $\hat{v}_\ell$ via (22) and (23);
  Calculate the estimated complex path gain $\hat{\beta}_\ell$ via (25);
end for
Classify P scatterer points with the largest velocities as moving targets and the rest $L-P$ as background scatterers according to velocity estimation $\hat{v}_\ell$, $\forall l \in [1, L]$;
Calculate the analog precoding matrix $\boldsymbol{F}_{\mathrm{RF}}$ and combining matrix $\boldsymbol{W}_{\mathrm{RF}}$ via (39);
Calculate the effective channel matrix $\dot{\boldsymbol{\Xi}}_{k,g}$ via (47);
Calculate the SVD of $\dot{\boldsymbol{\Xi}}_{k,g}$, and obtain digital precoding matrix $\boldsymbol{F}_{\mathrm{BB},k,g}$ and combining matrix $\boldsymbol{W}_{\mathrm{BB},k,g}$ via (46);

Output: $[\hat{x}_{\ell,1}, \hat{y}_{\ell,1}]^{\mathrm{T}}$, $\boldsymbol{F}_{\mathrm{RF}}$, $\boldsymbol{F}_{\mathrm{BB},k,g}$, $\boldsymbol{W}_{\mathrm{RF}}$, $\boldsymbol{W}_{\mathrm{BB},k,g}$.

rate without compromising precoding performance. The corresponding feedback rate can be calculated as $\frac{N_{\mathrm{s}} N_{\mathrm{p}} N_{\mathrm{b}}}{T_{\mathrm{f}}}$, where N_{b} is the number of quantization bits used for each parameter, N_{p} is the number of estimated parameters per scatterer, and T_{f} is the duration of feedback process.

Based on the analog and digital matrices given in (39) and (46), the hybrid precoding and combining matrices of the proposed scheme can be expressed as

$$\begin{aligned}\boldsymbol{F}_{\mathrm{pro},k,g}(m) &= \boldsymbol{F}_{\mathrm{RF}} \boldsymbol{F}_{\mathrm{BB},k,g}(m),\\ \boldsymbol{W}_{\mathrm{pro},k,g}(m) &= \boldsymbol{W}_{\mathrm{RF}} \boldsymbol{W}_{\mathrm{BB},k,g}(m).\end{aligned} \tag{48}$$

The procedure of the proposed sensing-aided precoding design is summarized in Algorithm 1.

3) Achievable Rate in High-Dynamic Scenarios: To further demonstrate the reliability of the proposed method in high-dynamic scenarios, we derive the achievable rate corresponding to the proposed precoding design and compare it to that of the optimal unconstrained design in (34). Specifically, by inserting (48) into (29), the equivalent channel matrix corresponding to the proposed precoding design is expressed as

$$\bar{\boldsymbol{H}}_{\mathrm{pro},k,g}(m) = [\boldsymbol{\Psi}_{k,g}]_{(\boldsymbol{\nu}_{(1:N_{\mathrm{s}})}, \boldsymbol{\nu}_{(1:N_{\mathrm{s}})})}. \tag{49}$$

Here, each diagonal element in $\bar{\boldsymbol{H}}_{\mathrm{pro},k,g}(m)$ corresponds to a sub-channel associated with a background scatterer. In other words, the columns of the precoding and combining matrices are orthogonal to the columns of the matrix in (43), which correspond to the mobile propagation paths. As such, the energy leakage over these paths is mitigated.

By substituting (49) into (28), the achievable rate corresponding to the proposed design can be derived as

$$\begin{aligned}&\mathcal{R}_{\mathrm{pro},g}(m)\\ &= \frac{1}{K}\sum_{k=1}^{K} \log_2\left(\left|\boldsymbol{I}_{N_{\mathrm{s}}} + \frac{\varrho}{N_{\mathrm{s}}\sigma_{\mathrm{n}}^2}\left(\boldsymbol{W}_{\mathrm{pro},k,g}^{\mathrm{H}} \boldsymbol{W}_{\mathrm{pro},k,g}\right)^{-1}\right.\right.\\ &\left.\left.\times [\boldsymbol{\Psi}_{k,g}]_{(\boldsymbol{\nu}_{(1:N_{\mathrm{s}})}, \boldsymbol{\nu}_{(1:N_{\mathrm{s}})})} [\boldsymbol{\Psi}_{k,g}]^{\mathrm{H}}_{(\boldsymbol{\nu}_{(1:N_{\mathrm{s}})}, \boldsymbol{\nu}_{(1:N_{\mathrm{s}})})}\right|\right).\end{aligned} \tag{50}$$

Different from the achievable rate corresponding to the optimal precoding design in (34), the above derivation does not depend on the assumptions of stationary scatterers, since the parameters within the equivalent channel matrix $\bar{\boldsymbol{H}}_{\mathrm{pro},k,g}(m)$ in (49) are solely associated with quasi-static propagation paths. As such, a higher resistibility to the mobility of scatterer points can be achieved, ensuring a more reliable transmission performance in high-dynamic scenarios.

V. Extension to Low-Resolution DAC Architecture

In practical communication infrastructure, DACs with theoretically infinite-bit resolution can lead to high power consumption and hardware complexity. In this regard, this section extends the proposed sensing-aided precoding framework to consider the utilization of one-bit DACs at the BS.[5]

Under the constraint of one-bit DACs, the transmitted signal $\boldsymbol{x}_{k,g}(m)$ in (5) can be reformulated as

$$\boldsymbol{x}_{k,g}(m) = \boldsymbol{F}_{\mathrm{RF}} \mathcal{Q}\left(\boldsymbol{F}_{\mathrm{BB},k,g}(m) \dot{\boldsymbol{x}}_{k,g}(m)\right), \tag{51}$$

where $\mathcal{Q}(\cdot)$ denotes the one-bit mapping function quantizing both real and imaginary components of its augments as

$$\mathcal{Q}(\cdot) = \mathrm{sgn}(\Re(\cdot)) + \mathrm{j}\,\mathrm{sgn}(\Im(\cdot)). \tag{52}$$

Based on the Bussgang's model [52], the one-bit quantification in (51) can be approximated by

$$\mathcal{Q}\left(\boldsymbol{F}_{\mathrm{BB},k,g}(m) \dot{\boldsymbol{x}}_{k,g}(m)\right) \approx \varsigma \boldsymbol{F}_{\mathrm{BB},k,g}(m) \dot{\boldsymbol{x}}_{k,g}(m) + \boldsymbol{\epsilon}(m), \tag{53}$$

where ς denotes the distortion constant for one-bit resolution, and $\boldsymbol{\epsilon}(m) \sim \mathcal{CN}\left(\boldsymbol{0}, \sigma_{\bar{\mathrm{n}}}^2 \boldsymbol{I}\right)$ denotes the additive quantization noise with variance $\sigma_{\bar{\mathrm{n}}}^2$.

Accordingly, when calculating the achievable rate $\mathcal{R}_g(m)$ in (28), the noise covariance matrix $\boldsymbol{R}_{\mathrm{n}}(m)$ should be replaced by the covariance matrix $\boldsymbol{R}_\eta(m)$ of the composite noise

$$\boldsymbol{\eta} = \boldsymbol{W}_{k,g}^{\mathrm{H}}(m) \boldsymbol{H}_{k,g}(m) \boldsymbol{F}_{\mathrm{RF}} \boldsymbol{\epsilon}(m) + \boldsymbol{W}_{k,g}^{\mathrm{H}}(m) \boldsymbol{n}_{k,g}(m), \tag{54}$$

which includes both Gaussian noise and quantization noise. As such, the modified noise covariance matrix is given by

$$\begin{aligned}\boldsymbol{R}_\eta(m) &= \sigma_{\bar{\mathrm{n}}}^2 \boldsymbol{W}_{k,g}^{\mathrm{H}}(m) \boldsymbol{H}_{k,g}(m) \boldsymbol{F}_{\mathrm{RF}} \boldsymbol{F}_{\mathrm{RF}}^{\mathrm{H}} \boldsymbol{H}_{k,g}^{\mathrm{H}}(m)\\ &\quad \times \boldsymbol{W}_{k,g}(m) + \sigma_{\mathrm{n}}^2 \boldsymbol{W}_{k,g}^{\mathrm{H}}(m) \boldsymbol{W}_{k,g}(m).\end{aligned} \tag{55}$$

The achievable rate $\tilde{\mathcal{R}}_g(m)$ under the constraint of one-bit DACs can then be written as

$$\begin{aligned}&\tilde{\mathcal{R}}_g(m)\\ &= \frac{1}{K}\sum_{k=1}^{K} \log_2\left(\left|\boldsymbol{I}_{N_{\mathrm{s}}} + \frac{\varrho}{N_{\mathrm{s}}} \boldsymbol{R}_\eta^{-1}(m) \bar{\boldsymbol{H}}_{k,g}(m) \bar{\boldsymbol{H}}_{k,g}^{\mathrm{H}}(m)\right|\right).\end{aligned} \tag{56}$$

[5]One-bit DACs are employed here as an example, and the signal model can be easily extended to more general cases of few-bit DACs. Other hardware impairments, such as phase noise and mutual coupling, can also degrade spectral efficiency. However, a detailed analysis and dedicated mitigation strategies for these impairments, particularly for one-bit hybrid system, are beyond the scope of this paper.

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

Importantly, the structure of $\boldsymbol{F}_{\mathrm{BB},k,g}(m)$ in (45) allows for the use of one-bit DACs at the BS without introducing quantization distortions in (51), i.e.,

$$\mathcal{Q}\left(\boldsymbol{F}_{\mathrm{BB},k,g}(m)\dot{\boldsymbol{x}}_{k,g}(m)\right) = \boldsymbol{F}_{\mathrm{BB},k,g}(m)\dot{\boldsymbol{x}}_{k,g}(m). \tag{57}$$

A detailed explanation of this statement is provided in the following remark.

Remark 2: According to *Theorem 2*, all the elements in $\tilde{\boldsymbol{V}}_{k,g}$ are restricted to the set $\{-1, 0, 1\}$, and thus the product $\boldsymbol{F}_{\mathrm{BB},k,g}(m)\dot{\boldsymbol{x}}_{k,g}(m)$ in (57) is confined to four values within the set $\{1+\mathrm{j}, 1-\mathrm{j}, -1+\mathrm{j}, -1-\mathrm{j}\}$ for QPSK modulation. As such, the one-bit mapping does not result in a quantization error of $\boldsymbol{F}_{\mathrm{BB},k,g}(m)\dot{\boldsymbol{x}}_{k,g}(m)$, relieving the need for additional iterative computations to compensate for the performance loss. For higher-order modulations, the minimum number of bits required in DACs for error-free quantization equals to the number of in-phase or quadrature-phase components per quadrant in the corresponding constellation diagram. Taking the 16-quadrature amplitude modulation (QAM) as an example, $\boldsymbol{F}_{\mathrm{BB},k,g}(m)\boldsymbol{s}_{k,g}(m)$ takes on the values $\{1+\mathrm{j}, 1+3\mathrm{j}, 3+\mathrm{j}, 3+3\mathrm{j}\}$ in the first quadrant, and the number of bits required in DACs equals to 2, which remains practical for deployment.

Therefore, the proposed method can be flexibly extended to higher-order modulation schemes with low-resolution DAC implementations, enabling energy-efficient precoding in diverse modulation scenarios.

VI. Analysis of Scatterer Identifiability and Computational Complexity

In this section, we discuss the scatterer identifiability based on the uniqueness condition for tensor decomposition and the computational complexity of the proposed method.

A. Scatterer Identifiability

To evaluate the upper limit of identifiable scatterers in the deployed system, we analyze the uniqueness of the optimization problem in (16). This ensures that the channel parameters associated with each distinct scatterer can be uniquely determined. Specifically, following the Kruskal's condition [53], the CP decomposition in (16) is unique as long as it satisfies

$$\kappa(\boldsymbol{B}_1) + \kappa(\boldsymbol{B}_2) + \kappa(\boldsymbol{B}_3) \geq 2L + 2, \tag{58}$$

where $\kappa(\boldsymbol{B}_1) = \min(M_{\mathrm{MS}}, L)$, $\kappa(\boldsymbol{B}_2) = \min(K, L)$, and $\kappa(\boldsymbol{B}_3) = \min(M, L)$. Normally, there are hundreds of fast-time samples in one PRI and tens of FMCW chirps within the duration of a subframe, making K and M much bigger than M_{MS}. When $L > \max(K, M)$, we can substitute $\kappa(\boldsymbol{B}_1) = M_{\mathrm{MS}}$, $\kappa(\boldsymbol{B}_2) = K$, and $\kappa(\boldsymbol{B}_3) = M$ into (58), leading to $\max(K, M) < L \leq (M_{\mathrm{MS}} + K + M - 2)/2$. Likewise, when $\min(K, M) < L \leq \max(K, M)$, we have $\min(K, M) < L \leq \min\{\min(K, M) + M_{\mathrm{MS}} - 2, \max(K, M)\}$. Moreover, when $L \leq \min(K, M)$, the Kruskal's condition remains valid since $M_{\mathrm{MS}} \geq 2$ holds true for hybrid systems. However, in practical applications, the scatterer identifiability is subject to other constraints, such as the size of antenna arrays, the system bandwidth, the gains in power at the BS and the MS, and the radar cross-section (RCS) of the scatterers. Due to these restrictions, the actual number of identifiable scatterers can be smaller than the maximum theoretical value derived from the uniqueness condition for CP decomposition.

B. Computational Complexity

The computational complexity of the proposed method is associated with channel estimation and precoding design. For the part of channel estimation, the computational complexity of calculating the factor matrices in (16) can be measured by $\mathcal{O}(M_{\mathrm{MS}}KMLN_{\mathrm{ALS}})$, where N_{ALS} denotes the number of iterations in the ALS algorithm. As for the proposed precoding design, the digital precoding and combining matrices are calculated through SVD of the effective channel matrix $\dot{\boldsymbol{\Xi}}_{k,g}(m)$ in (47) with the computational complexity of $\mathcal{O}(N_{\mathrm{s}}^3 K)$.

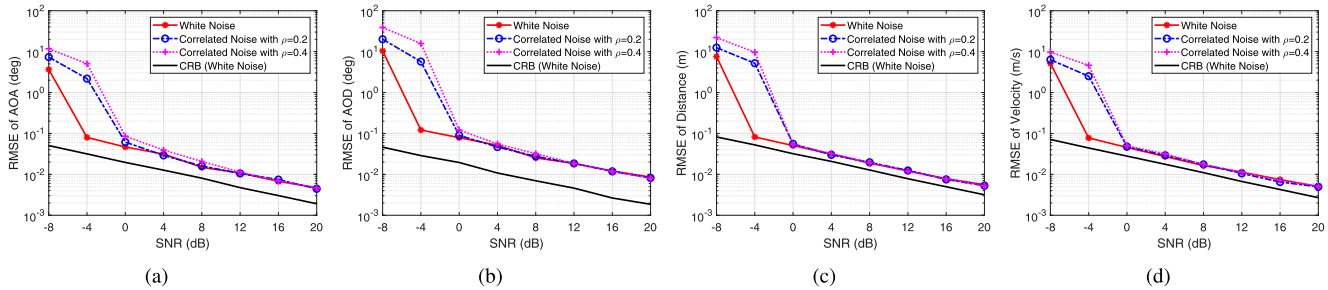
In comparison, the conventional precoding designs involve computing the SVD of the full channel matrix $\tilde{\boldsymbol{H}}_{k,g}(m)$ as expressed in (33) and iteratively calculating the precoding and combining matrices. For instance, the computational complexities of the orthogonal matching pursuit (OMP)-based method [54] and the manifold-optimization (MO)-based method [55] are respectively $\mathcal{O}(N_{\mathrm{BS}}N_{\mathrm{MS}}^2 K + N_{\mathrm{iter}}N_{\mathrm{BS}}M_{\mathrm{MS}}N_{\mathrm{s}}LK)$ and $\mathcal{O}(N_{\mathrm{BS}}N_{\mathrm{MS}}^2 K + N_{\mathrm{iter}}N_{\mathrm{BS}}^2 M_{\mathrm{MS}}K)$, where N_{iter} denotes the number of iterations to achieve convergence. Note that, these computational complexities exclude that of channel estimation.

Given the limited number of independent data streams N_{s} compared to the number of array elements N_{BS} and N_{MS}, the proposed method is computationally more efficient than the conventional methods, making it well-suited for real-world implementations, as will also be confirmed in the next section.

VII. Numerical Results

In this section, we present numerical results to evaluate both the communication and sensing performance of the proposed sensing-aided precoding method. System configurations conforming to the 3GPP Specification [56] are specified in Table II. For the mmWave channel, the number of propagation paths is set to $L = 6$. Unless otherwise stated, $P = 4$ paths correspond to fast-moving targets, and the rest $Q = 2$ paths correspond to background scatterers. The AOAs and AODs are randomly distributed in $[-\pi/3, \pi/3]$. The velocities of the fast-moving targets are randomly generated from the range $[20, 30]$ m/s, while the quasi-static background scatterers have velocities randomly generated from the interval $[0, 2]$ m/s. The complex path gains are uniformly chosen from the unit circle in the complex plane. The distance between the MS and BS is $d_{\mathrm{bis}} = 100$ m. During the channel estimation stage, the BS applies beamforming vectors as described in Subsection II-B, sweeping through a uniform angular grid from $-\pi/3$ to $\pi/3$. The combining matrices at the MS, $\tilde{\boldsymbol{W}}_{\mathrm{RF}}$ and $\tilde{\boldsymbol{W}}_{\mathrm{BB}}$, are randomly generated while satisfying the constraints in (30). During data transmission, the number of data streams is set to $N_{\mathrm{s}} = 2$. The simulation is conducted using MATLAB on a desktop PC equipped with an Intel(R) Core(TM) i7-12700 CPU and 32 GB RAM.

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d)

Fig. 6. Parameter estimation accuracy in the first subframe under different noise conditions. (a) RMSE of AOA $\hat{\theta}_{\ell,1}$. (b) RMSE of AOD $\hat{\phi}_{\ell,1}$. (c) RMSE of distance $\hat{d}_{\ell,1}$. (d) RMSE of velocity $\hat{v}_{\ell}$. For each data point of the curves, $J = 500$ Monte-Carlo trials are used. In addition, the Cramér–Rao bounds (CRBs) for the estimation of different parameters in the case of white noise are shown for reference.

TABLE II
SYSTEM PARAMETERS

Parameter	Symbol	Value
Number of antennas at BS	N_{BS}	64
Number of antennas at MS	N_{MS}	32
Number of RF chains at BS	M_{BS}	8
Number of RF chains at MS	M_{MS}	4
Antenna spacing	d	0.5λ
Carrier frequency (GHz)	f_c	28
Subcarrier spacing (kHz)	Δ	120
Signal bandwidth (MHz)	B	30.72
Number of subframes per time frame	G	40
Number of subcarriers for OFDM / fast-time samples for FMCW	K	256
Number of OFDM symbols / FMCW chirps in a subframe	M	112
OFDM symbol duration (μs)	T_{s}	8.92
FMCW pulse repetition interval (μs)	T_{r}	8.92
FMCW chirp duration (μs)	T_0	8.33

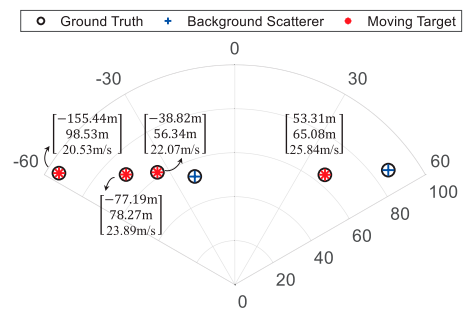


Fig. 7. Moving target identification and localization, SNR $= 0$ dB. The results of localization and velocity estimation are represented in the form of $[\hat{x}_{\ell,1}, \hat{y}_{\ell,1}, \hat{v}_{\ell}]^{\mathrm{T}}$.

A. Moving Target Identification and Localization

To validate the sensing functionalities in the proposed scheme, the performance of moving target identification and localization is illustrated in Fig. 7. It is observed that both the moving targets and quasi-static scatterers can be distinguished and accurately localized. The accuracy of the location channel parameters $\{\theta_{\ell,1}, \phi_{\ell,1}, d_{\ell,1}, v_{\ell}\}$ estimated in the first subframe are shown in Fig. 6, where the root-mean-square error (RMSE) is utilized as the performance metric. To evaluate the robustness of parameter estimation under different noise conditions, we consider both cases of white noise and spatially correlated noise, where the correlation coefficient of the noise across various data streams ρ is set to $\rho = 0.2$ and $\rho = 0.4$, respectively. As depicted in Fig. 6, the proposed method exhibits reliable performance across all investigated noise scenarios for SNR values above 0 dB. The convergence of the RMSE curves underscores its resilience to spatial noise correlation in realistic channel environments.

B. Comparison of Achievable Rate

In this subsection, the achievable rate of the proposed method is compared to those of the OMP-based method ('OMP') [54], the MO-based method ('MO') [55], the OMP-based method with additional channel tracking ('OMP-CT') [33], [34], the average minimum mean square error (MMSE)-based method ('Avg-MMSE') [57], and the maximin-based method ('Maximin') [58]. These methods were selected as baselines because they are well-established, model-driven approaches widely used for hybrid precoding in mmWave MIMO systems, providing a relevant benchmark for the proposed method. In the case with channel tracking, the channel parameters are estimated again in the 21st subframe by sending preambles in the null-space of the estimated channel matrix to prevent interfering with data transmission. Then, the estimated parameters $\{\hat{\theta}_{\ell,21}, \hat{\phi}_{\ell,21}, \hat{d}_{\ell,21}\}$ are utilized for precoding design in the subsequent subframes. The maximum number of iterations for MO-based method is set to 10. The average MMSE-based method minimizes the average mean square error under statistical CSI uncertainty, where the channel error variance is set to 0.05 in our simulations. The maximin-based method optimizes the worst-case performance under deterministic norm-bounded channel errors by solving a min-max problem, where the uncertainty bound is set to 1.

1) Comparison Under Different Configurations of DACs: In Fig. 8, the achievable rate of the tested methods are compared under infinite-bit DACs. It is clear that the achievable rate of the proposed method succeeds those of competing methods. The reason is that the differences between the estimated channel matrix $\hat{\boldsymbol{H}}_{k,g}$ in (26) and the actual channel matrix $\boldsymbol{H}_{k,g}$ in (13) may result in interferences between data streams corresponding to the optimal design in (32). Since the OMP-based and MO-based methods rely on the results of the optimal design, their

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

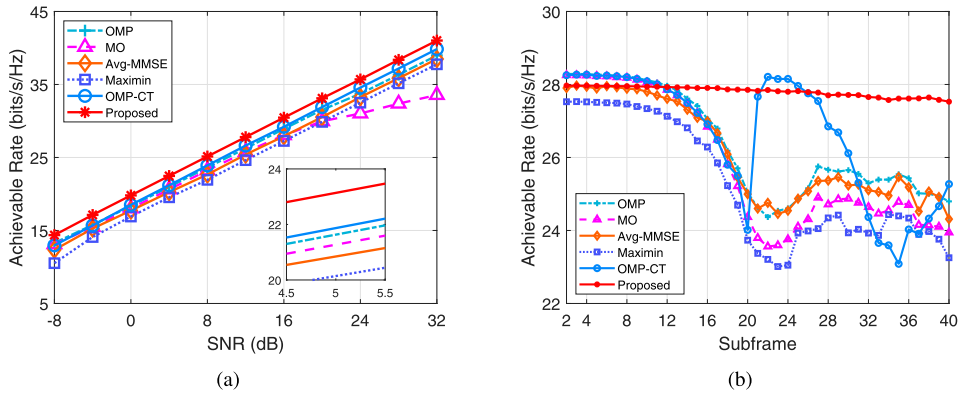


Fig. 8. Comparison of achievable rates with infinite-bit DACs. (a) Achievable rate *versus* SNR, where the achievable rate is averaged from the 2nd subframe to the 40th subframe. (b) Achievable rate *versus* subframe index, where SNR $= 12$ dB.

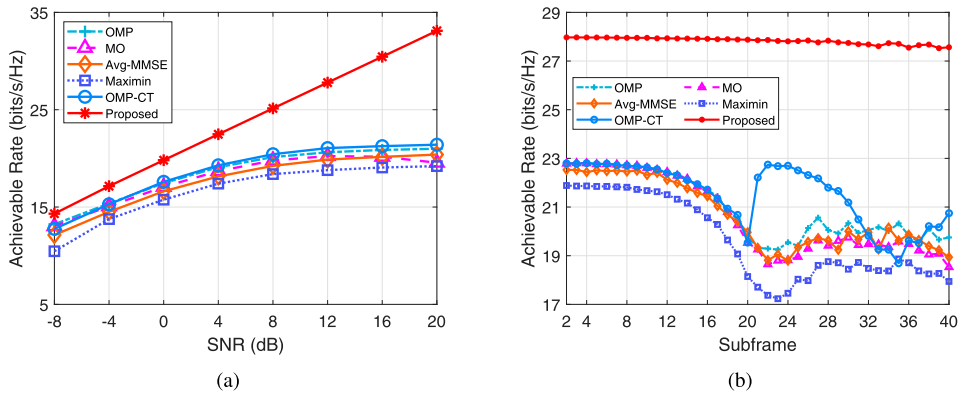


Fig. 9. Comparison of achievable rates with one-bit DACs. (a) Achievable rate *versus* SNR, where the achievable rate is averaged from the 2nd subframe to the 40th subframe. (b) Achievable rate *versus* subframe index, where SNR $= 12$ dB.

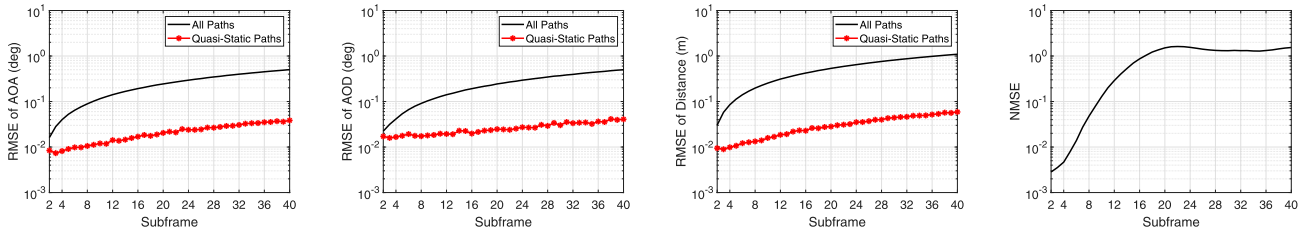
effectiveness is inevitably deteriorated. Moreover, the MO-based method deviates from a linear increment because of its sensitivity to initialization and objective function saturation at high SNR values. Although the average MMSE-based method and maximin-based method are designed for robust precoding, their performance is limited in high-dynamic scenarios considered in this paper. Particularly, the average MMSE-based method assumes that channel estimation errors follow a specific statistical distribution. However, errors induced by scatterer mobility may violate this assumption because the underlying channel variations exhibit spatial and temporal correlations. Meanwhile, the maximin-based method focuses on optimizing worst-case performance, which compromises average achievable rates.

Furthermore, the time-varying achievable rates corresponding to the methods tested are demonstrated in Fig. 8(b). Without channel tracking, the achievable rates of the competing methods decrease steadily during the first half of a frame, then maintain a relatively low level with fluctuations. These trends in achievable rates correlate with the estimation error in the channel matrix, as will be demonstrated in Subection VII-C. Due to additional channel estimation procedure, the OMP-based method with channel tracking exhibits a notable increase in achievable rate during the 21st subframe. In contrast, the proposed method shows stable achievable rates throughout the entire frame without relying on channel tracking.

Then, the performance of the tested methods is simulated using one-bit DACs in Fig. 9. It is clear that the proposed method outperforms other methods with one-bit DACs in terms of achievable rate. This is because the digitally precoded symbols take on only one value in both the real and imaginary parts, thereby preventing the quantization error induced by one-bit mapping, as discussed in *Remark 2* of Section V. In contrast, when imposing one-bit DACs to the methods compared, extra distortion is introduced and the achievable rates are thereby degraded.

2) Comparison Versus Number of Antennas at the MS: In Fig. 11, we compare the achievable rates among all the methods evaluated versus number of antennas N_{MS} at the MS. It can be observed that the proposed method maintains its effectiveness across varying N_{MS}. This robustness stems from capability of the proposed method to suppress energy leakage from fast-moving scatterers in both the analog and digital domains. In particular, even if the orthogonality in the analog domain between paths of quasi-static scatterers and moving targets degrades with fewer antennas, the orthogonality in the digital domain is preserved in the distinct columns of the unitary matrices (see (40)) corresponding to these different types of scatterers. This

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d)

Fig. 10. Parameter and channel estimation error *versus* subframe index, SNR $= 12$ dB. (a) RMSE of AOA $\hat{\theta}_{\ell,1}$. (b) RMSE of AOD $\hat{\phi}_{\ell,1}$. (c) RMSE of distance $\hat{d}_{\ell,1}$. (d) NMSE of channel matrix $\hat{\boldsymbol{H}}_{k,g}$.

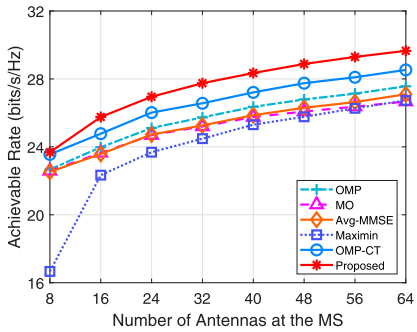


Fig. 11. Comparison of averaged achievable rates versus number of antennas at the MS, where SNR $= 12$ dB.

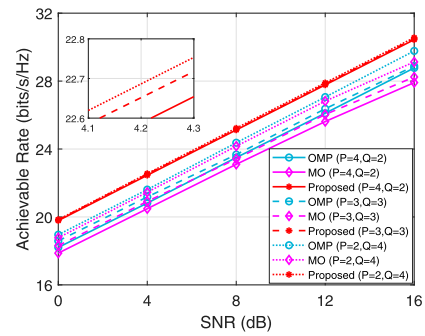


Fig. 12. Comparison of averaged achievable rates versus number of quasi-static scatterers and moving targets.

ensures the effectiveness of the proposed method, even for the case of MSs with limited number of antennas.

3) Comparison Versus Number of Quasi-static and Mobile Scatterers: Fig. 12 compares the achievable rates of the proposed method against the OMP-based method and the MO-based method across varying proportions of quasi-static and mobile scatterers. In Fig. 12, the proposed method consistently outperforms these benchmarks, yielding higher achievable rates across all evaluated conditions. Notably, the performance superiority of the proposed method becomes increasingly significant as the proportion of mobile scatterers increases. This is attributed to the ability of the proposed method to deal with the increased temporal dynamics within the channel matrix. Specifically, it mitigates mobility-induced fluctuations by leveraging precoding and combining matrices (48) designed to concentrate energy onto the quasi-static propagation paths, as defined in (49). In contrast, conventional methods that rely on accurate instantaneous CSI estimation become less effective under such variations.

C. Effectiveness of Proposed Channel Division Criterion

To illustrate the reasons for the decline in achievable rates of the methods compared in Fig. 8(b), the averaged RMSEs of time-varying parameters for all the propagation paths and only the quasi-static paths are demonstrated in Fig. 10. Over time, the averaged difference between the location parameters estimated in the first subframe $\{\hat{\theta}_{\ell,1}, \hat{\phi}_{\ell,1}, \hat{d}_{\ell,1}\}$ and the actual parameters in the g-th subframe $\{\theta_{\ell,g}, \phi_{\ell,g}, d_{\ell,g}\}$ increases for all propagation paths. This growing parameter error leads to discrepancies between the estimated channel matrix $\hat{\boldsymbol{H}}_{k,g}$ in (26) and actual channel matrix $\boldsymbol{H}_{k,g}$ in (13), characterized by normalized mean square error (NMSE) defined as

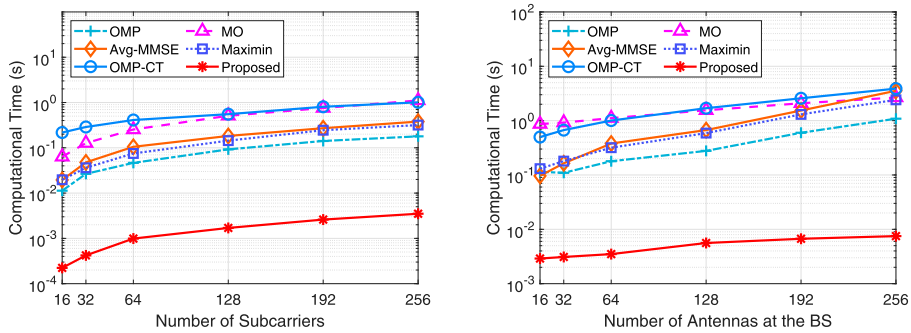
$$\mathrm{NMSE} = \frac{1}{JK}\sum_{j=1}^{J}\sum_{k=1}^{K}\frac{\left\|\{\boldsymbol{H}_{k,g}\}_j - \{\hat{\boldsymbol{H}}_{k,g}\}_j\right\|_{\mathrm{F}}^2}{\|\{\boldsymbol{H}_{k,g}\}_j\|_{\mathrm{F}}^2}, \tag{59}$$

where $\{\boldsymbol{H}_{k,g}\}_j$ denotes the channel matrix corresponding to the k-th subcarrier and g-th subframe in the j-th Monte-Carlo trial, and $\{\hat{\boldsymbol{H}}_{k,g}\}_j$ represents its estimation. It can be observed from Fig. 10(d) that the NMSE of the channel estimate increases rapidly for the subframes distant from the first subframe, following similar trend to that of the parameter estimation error. In contrast, the estimation errors of the parameters of the quasi-static propagation paths remain insignificant over time as shown in Fig. 10(a)–10(c), thereby guaranteeing reliable communication performance.

D. Comparison of Computation Time

Finally, we compare in Fig. 13 the computational complexities of the evaluated methods by demonstrating their computation time averaged from that of each symbol duration within the 21st subframe. In Fig. 13(a), the number of subcarriers K varies while other settings are fixed. It is clear that the proposed method has the lowest computation time among all the methods tested, since it does not include any complex optimization process to approximate the optimal solutions as characterized in (31). Furthermore, we compare the computation time of the

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 13. Comparison of computational complexity. (a) Computation time *versus* the number of subcarriers K, where $N_{\mathrm{BS}} = 64$. (b) Computation time *versus* the number of antennas N_{BS}, where $K = 256$.

methods tested when the number of antennas at the BS N_{BS} varies, as shown in Fig. 13(b). Since the computations needed for the proposed method scale with the number of independent data streams rather than the number of antennas, the proposed method tends to be computational more efficient with large antenna arrays, which have been widely deployed in massive MIMO systems. It makes the proposed method well-suited for real-world implementations.

VIII. Conclusion & Discussion

In this paper, we have proposed a sensing-aided precoding method for hybrid MIMO systems, which achieves enhanced spectral efficiency in high-dynamic scenarios. Specifically, we investigated the variations of time-varying channel parameters and established a channel division criterion to partition the channel matrix into two components representing mobile and quasi-static propagation paths, respectively. By exploiting the digital-analog dual orthogonality within this divided channel matrix, the precoding and combining matrices were designed to concentrate data transmission predominantly onto the quasi-static paths in both analog and digital domains. As such, the adverse impacts of moving targets were mitigated, leading to improved communication performance in highly dynamic environments. Simulation results validated the superior performance of the proposed method in terms of achievable rate and computational complexity.

Beyond the current bistatic framework, the core principles of our method can be adapted to monostatic and multistatic configurations. In a monostatic setup, location parameters can still be effectively estimated from FMCW echoes, while dedicated communication pilots are necessary for complex path gain estimation. In distributed configurations with multiple receiving MSs monitoring the same region, the BS establishes a bistatic link with each MS. The proposed parameter estimation scheme is then applied independently for each link. Crucially, such multistatic configuration can resolve the inherent ambiguity associated with bistatic velocity estimation through observations from multiple perspectives. Further development of precoding methods leveraging this enhanced environmental awareness underscores the potential for more advanced and intelligent sensing-aided communication systems.

Appendix A
Proof of Theorem 1

According to the relationship between SVD and eigenvalue decomposition, we have

$$\begin{aligned}\hat{\boldsymbol{\Xi}}_{k,g}^{\mathrm{H}}\hat{\boldsymbol{\Xi}}_{k,g} &= \tilde{\boldsymbol{V}}_{k,g}\boldsymbol{\Psi}_{k,g}^{\mathrm{H}}\tilde{\boldsymbol{U}}_{k,g}^{\mathrm{H}}\tilde{\boldsymbol{U}}_{k,g}\boldsymbol{\Psi}_{k,g}\tilde{\boldsymbol{V}}_{k,g}^{\mathrm{H}}\\ &= \tilde{\boldsymbol{V}}_{k,g}(\boldsymbol{\Psi}_{k,g}^{\mathrm{H}}\boldsymbol{\Psi}_{k,g})\tilde{\boldsymbol{V}}_{k,g}^{\mathrm{H}}.\end{aligned} \tag{60}$$

Herein, both $\boldsymbol{\Xi}_{k,g}^{\mathrm{H}}\boldsymbol{\Xi}_{k,g}$ and $\boldsymbol{\Psi}_{k,g}^{\mathrm{H}}\boldsymbol{\Psi}_{k,g}$ are positive semidefinite diagonal matrices. Therefore, according to Cholesky decomposition theorem, there is one unique positive semidefinite diagonal matrix $\bar{\boldsymbol{\Psi}}_{k,g}$ such that $\boldsymbol{\Xi}_{k,g}^{\mathrm{H}}\boldsymbol{\Xi}_{k,g} = \bar{\boldsymbol{\Psi}}_{k,g}^{\mathrm{H}}\bar{\boldsymbol{\Psi}}_{k,g}$. By letting $\boldsymbol{\Psi}_{k,g} = \bar{\boldsymbol{\Psi}}_{k,g}$, $\tilde{\boldsymbol{V}}_{k,g}$ can be any diagonal matrix with -1 or 1 on its diagonal. Meanwhile, the SVD of $\boldsymbol{\Xi}$ in (40) is unique up to the permutation ambiguities in columns of $\tilde{\boldsymbol{V}}_{k,g}$. When the diagonal elements of $\boldsymbol{\Psi}_{k,g}$ interchange their positions, the corresponding elementary matrix transformations are performed on $\tilde{\boldsymbol{V}}_{k,g}$. ■

Appendix B
Proof of Theorem 2

According to the definition of $\boldsymbol{\nu}_{(p)}$ in (41), we have the following equations

$$\begin{aligned}[\tilde{\boldsymbol{U}}_{k,g}]_{(\cdot,\boldsymbol{\nu}_{(p)})}[\tilde{\boldsymbol{U}}_{k,g}]_{(\cdot,\boldsymbol{\nu}_{(p)})}^{\mathrm{H}} &= \mathrm{diag}(\boldsymbol{w}_p),\\ [\tilde{\boldsymbol{V}}_{k,g}]_{(\cdot,\boldsymbol{\nu}_{(p)})}[\tilde{\boldsymbol{V}}_{k,g}]_{(\cdot,\boldsymbol{\nu}_{(p)})}^{\mathrm{H}} &= \mathrm{diag}(\boldsymbol{w}_p),\end{aligned} \tag{61}$$

where $\boldsymbol{w}_p \in \mathbb{N}^L$ is a vector with 1 on its p-th entry and 0 on all other entries. For the p-th moving target, the expression in the first line of (43) can be derived as

$$\begin{aligned}&\boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})[\tilde{\boldsymbol{U}}_{k,g}]_{(\cdot,\boldsymbol{\nu}_{(p)})}[\tilde{\boldsymbol{\Psi}}_{k,g}]_{(\boldsymbol{\nu}_{(p)},\boldsymbol{\nu}_{(p)})}[\tilde{\boldsymbol{V}}_{k,g}]_{(\cdot,\boldsymbol{\nu}_{(p)})}^{\mathrm{H}}\boldsymbol{A}_{\mathrm{BS}}^{\mathrm{H}}(\hat{\boldsymbol{\phi}})\\ &= \boldsymbol{A}_{\mathrm{MS}}(\hat{\boldsymbol{\theta}})\mathrm{diag}(\boldsymbol{w}_p)[\hat{\boldsymbol{\Xi}}_{k,g}]_{(\boldsymbol{\nu}_{(p)},\boldsymbol{\nu}_{(p)})}\mathrm{diag}(\boldsymbol{w}_p)\boldsymbol{A}_{\mathrm{BS}}^{\mathrm{H}}(\hat{\boldsymbol{\phi}})\\ &= \boldsymbol{a}_{\mathrm{MS}}(\hat{\theta}_{p,1})\hat{\zeta}_{p,k,g}\boldsymbol{a}_{\mathrm{BS}}(\hat{\phi}_{p,1}).\end{aligned} \tag{62}$$

By performing the above calculation for $p = 1, 2, \cdots, P$ and gathering the derivations into a compact form, the equation (43) can be obtained. Similarly, for the background scatterers with index $q = P+1, P+2, \cdots, L$, the equation (44) can be obtained by substituting p with q, and $\boldsymbol{\nu}$ with $\acute{\boldsymbol{\nu}}$ in (62). ■

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] L. Lin, H. Zheng, S. A. Vorobyov, C. Zhou, and Z. Shi, "Sensing-aided communication channel estimation with tensor-based moving target localization," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, Seoul, South Korea, Apr. 2024, pp. 8606–8610.

[2] "Framework and overall objectives of the future development of IMT for 2030 and beyond," Int. Telecommun. Union (ITU), Geneva, Switzerland, Tech. Rep. Recommendation ITU-R M.2160-0, Nov. 2023.

[3] N. González-Prelcic et al., "The integrated sensing and communication revolution for 6G: Vision, techniques, and applications," *Proc. IEEE*, vol. 112, no. 7, pp. 676–723, Jul. 2024.

[4] Y. Wang, M. S. Obaidat, Y. Yin, L. Wang, J. J. P. C. Rodrigues, and B. Sadoun, "Robust sparse direct localization of smart vehicle with partly calibrated time modulated arrays," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 11, pp. 12 582–12 596, Nov. 2023.

[5] Y. Tian, G. Pan, and M.-S. Alouini, "Applying deep-learning-based computer vision to wireless communications: Methodologies, opportunities, and challenges," *IEEE Open J. Commun. Soc.*, vol. 2, pp. 132–143, 2020.

[6] S. Kim, J. Moon, J. Wu, B. Shim, and M. Z. Win, "Vision-aided positioning and beam focusing for 6G terahertz communications," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 9, pp. 2503–2519, Sep. 2024.

[7] F. Wen, W. Xu, F. Gao, C. Pan, and G. Liu, "Vision aided environment semantics extraction and its application in mmWave beam selection," *IEEE Commun. Lett.*, vol. 27, no. 7, pp. 1894–1898, Jul. 2023.

[8] J. Morais, A. Bchboodi, H. Pezeshki, and A. Alkhateeb, "Position-aided beam prediction in the real world: How useful GPS locations actually are?" in *Proc. IEEE Int. Conf. Commun. (ICC)*, Rome, Italy, May 2023, pp. 1824–1829.

[9] V. Va, J. Choi, T. Shimizu, G. Bansal, and R. W. Heath Jr., "Inverse multipath fingerprinting for millimeter wave V2I beam alignment," *IEEE Trans. Veh. Technol.*, vol. 67, no. 5, pp. 4042–4058, May 2018.

[10] A. Ali, N. González-Prelcic, and A. Ghosh, "Passive radar at the roadside unit to configure millimeter wave vehicle-to-infrastructure links," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 14903–14917, Dec. 2020.

[11] U. Demirhan and A. Alkhateeb, "Radar aided 6G beam prediction: Deep learning algorithms and real-world demonstration," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Austin, TX, USA, Apr. 2022, pp. 2655–2660.

[12] N. González-Prelcic, R. Méndez-Rial, and R. W. Heath Jr., "Radar aided beam alignment in mmWave V2I communications supporting antenna diversity," in *Proc. Inf. Theory Appl. Workshop (ITA)*, La Jolla, CA, USA, Feb. 2016, pp. 1–7.

[13] Y. Zhang, K. Patel, S. Shakkottai, and R. W. Heath Jr., "Side-information-aided noncoherent beam alignment design for millimeter wave systems," in *Proc. ACM Int. Symp. Mobile Ad Hoc Netw. Comput. (MobiHoc)*, Catania, Italy, Jul. 2019, pp. 341–350,

[14] A. Graff, Y. Chen, N. González-Prelcic, and T. Shimizu, "Deep learning-based link configuration for radar-aided multiuser mmWave vehicle-to-infrastructure communication," *IEEE Trans. Veh. Technol.*, vol. 72, no. 6, pp. 7454–7468, Jun. 2023.

[15] M. B. Mashhadi, M. Jankowski, T.-Y. Tung, S. Kobus, and D. Gündüz, "Federated mmWave beam selection utilizing LIDAR data," *IEEE Wireless Commun. Lett.*, vol. 10, no. 10, pp. 2269–2273, Oct. 2021.

[16] A. Klautau, N. González-Prelcic, and R. W. Heath Jr., "LiDAR data for deep learning-based mmWave beam-selection," *IEEE Wireless Commun. Lett.*, vol. 8, no. 3, pp. 909–912, Jun. 2019.

[17] D. Marasinghe et al., "LiDAR aided wireless networks - Beam prediction for 5G," in *Proc. IEEE Veh. Technol. Conf. (VTC)*, London, U.K., Sep. 2022, pp. 1–7.

[18] T. Woodford, X. Zhang, E. Chai, K. Sundaresan, and A. Khojastepour, "SpaceBeam: Lidar-driven one-shot mmWave beam management," in *Proc. ACM Int. Conf. Mobile Syst. Appl. Serv. (MobiSys)*, Virtual Event, WI, USA, Jun. 2021, pp. 389–401.

[19] A. Ali, N. González-Prelcic, R. W. Heath Jr., and A. Ghosh, "Leveraging sensing at the infrastructure for mmWave communication," *IEEE Commun. Mag.*, vol. 58, no. 7, pp. 84–89, Jul. 2020.

[20] K. V. Mishra, M. Bhavani Shankar, V. Koivunen, B. Ottersten, and S. A. Vorobyov, "Toward millimeter-wave joint radar communications: A signal processing perspective," *IEEE Signal Process. Mag.*, vol. 36, no. 5, pp. 100–114, Sep. 2019.

[21] P. Kumari, J. Choi, N. González-Prelcic, and R. W. Heath Jr., "IEEE 802.11ad-based radar: An approach to joint vehicular communication-radar system," *IEEE Trans. Veh. Technol.*, vol. 67, no. 4, pp. 3012–3027, Apr. 2018.

[22] P. Kumari, S. A. Vorobyov, and R. W. Heath Jr., "Adaptive virtual waveform design for millimeter-wave joint communication–radar," *IEEE Trans. Signal Process.*, vol. 68, pp. 715–730, 2020.

[23] N. Garcia, H. Wymeersch, E. G. Ström, and D. Slock, "Location-aided mm-wave channel estimation for vehicular communication," in *Proc. IEEE Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, Edinburgh, U.K., Jul. 2016, pp. 1–5.

[24] F. Liu, W. Yuan, C. Masouros, and J. Yuan, "Radar-assisted predictive beamforming for vehicular links: Communication served by sensing," *IEEE Trans. Wireless Commun.*, vol. 19, no. 11, pp. 7704–7719, Nov. 2020.

[25] N. González-Prelcic, A. Ali, V. Va, and R. W. Heath Jr., "Millimeter-wave communication with out-of-band information," *IEEE Commun. Mag.*, vol. 55, no. 12, pp. 140–146, Dec. 2017.

[26] S. Mura, M. Mizmizi, U. Spagnolini, and A. Petropulu, "Enhanced channel estimation in mm-Wave MIMO systems leveraging integrated communication and sensing," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, Seoul, South Korea, Apr. 2024, pp. 13101–13105.

[27] Z. Huang, K. Wang, A. Liu, Y. Cai, R. Du, and T. X. Han, "Joint pilot optimization, target detection and channel estimation for integrated sensing and communication systems," *IEEE Trans. Wireless Commun.*, vol. 21, no. 12, pp. 10351–10365, Dec. 2022.

[28] M. Dias, A. Klautau, N. González-Prelcic, and R. W. Heath Jr., "Position and LiDAR-aided mmWave beam selection using deep learning," in *Proc. IEEE Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, Cannes, France, Jul. 2019, pp. 1–5.

[29] B. Salehi et al., "Deep learning on multimodal sensor data at the wireless edge for vehicular network," *IEEE Trans. Veh. Technol.*, vol. 71, no. 7, pp. 7639–7655, Jul. 2022.

[30] G. Charan, T. Osman, A. Hredzak, N. Thawdar, and A. Alkhateeb, "Vision-position multi-modal beam prediction using real millimeter wave datasets," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Austin, TX, USA, Apr. 2022, pp. 2727–2731.

[31] D. Roy et al., "Going beyond RF: A survey on how AI-enabled multimodal beamforming will shape the nextG standard," *Comput. Netw.*, vol. 228, Jun. 2023, Art. no. 109729.

[32] K. Patel and R. W. Heath, "Harnessing multimodal sensing for multi-user beamforming in mmWave system," *IEEE Trans. Wireless Commun.*, vol. 23, no. 12, pp. 18725–18739, Dec. 2024.

[33] K. Upadhya, R. W. Heath Jr., and S. A. Vorobyov, "Tracking abruptly changing channels in mmWave systems using overlaid data and training," in *Proc. IEEE Int. Workshop Comput. Adv. Multi-Sens. Adapt. Process. (CAMSAP)*, Curacao, The Netherlands, Dec. 2017, pp. 1–5.

[34] K. Upadhya, S. A. Vorobyov, and R. W. Heath Jr., "Low-overhead receiver-side channel tracking for mmWave MIMO," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP),* Calgary, AB, Canada, Apr. 2018, pp. 3859–3863.

[35] M. Li, S. Zhang, Y. Ge, Z. Li, F. Gao, and P. Fan, "STAR-RIS aided integrated sensing and communication over high mobility scenario," *IEEE Trans. Commun.*, vol. 72, no. 8, pp. 4788–4802, Aug. 2024.

[36] L. Lin, J. Tong, H. Lin, H. Zheng, and Z. Shi, "Joint target localization and channel estimation for ODDM-ISAC systems," *IEEE Signal Process. Lett.*, vol. 32, pp. 1525–1529, 2025.

[37] Z. Zhou, J. Fang, L. Yang, H. Li, Z. Chen, and R. S. Blum, "Low-rank tensor decomposition-aided channel estimation for millimeter wave MIMO-OFDM systems," *IEEE J. Sel. Areas Commun.*, vol. 35, no. 7, pp. 1524–1538, Jul. 2017.

[38] F. Guidi, A. Guerra, and D. Dardari, "Personal mobile radars with millimeter-wave massive arrays for indoor mapping," *IEEE Trans. Mobile Comput.*, vol. 15, no. 6, pp. 1471–1484, Jun. 2016.

[39] X. Ma, Z. Gao, F. Gao, and M. Di Renzo, "Model-driven deep learning based channel estimation and feedback for millimeter-wave massive hybrid MIMO systems," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 8, pp. 2388–2406, Aug. 2021.

[40] M. L. Rahman, J. A. Zhang, X. Huang, Y. J. Guo, and R. W. Heath Jr., "Framework for a perceptive mobile network using joint communication and radar sensing," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 56, no. 3, pp. 1926–1941, Jun. 2020.

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

[41] Z. Xu and A. Petropulu, "A bandwidth efficient dual-function radar communication system based on a MIMO radar using OFDM waveforms," *IEEE Trans. Signal Process.*, vol. 71, pp. 401–416, Feb. 2023.
[42] S. Xu and A. Yarovoy, "Joint features extraction for multiple moving targets using (ultra-)wideband FMCW signals in the presence of Doppler ambiguity," *IEEE Trans. Signal Process.*, vol. 68, pp. 6562–6577, 2020.
[43] "Study on channel model for frequencies from 0.5 to 100 GHz," 3rd Generation Partnership Project (3GPP), version 18.0.0, Tech. Rep. (TR) 38.901, Mar. 2024.
[44] Y. Wang et al., "Tensor-based sparsity-inducing localization of AAV swarms-assisted mobile edge computing systems," *IEEE Trans. Ind. Informat.*, vol. 21, no. 3, pp. 2074–2083, Mar. 2025.
[45] F. Xu, H. Zheng, and S. A. Vorobyov, "Tensor-based 2D DOA estimation for L-shaped nested array," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 1, pp. 604–618, Feb. 2024.
[46] P. Comon, X. Luciani, and A. L. F. de Almeida, "Tensor decompositions, alternating least squares and other tales," *J. Chemom*,, vol. 23, no. 7/8, pp. 393–405, Apr. 2009.
[47] H. Zheng, C. Zhou, Z. Shi, and Y. Gu, "Structured tensor reconstruction for coherent DOA estimation," *IEEE Signal Process. Lett.*, vol. 29, pp. 1634–1638, 2022.
[48] H. Zheng, C. Zhou, Z. Shi, Y. Gu, and Y. D. Zhang, "Coarray tensor direction-of-arrival estimation," *IEEE Trans. Signal Process.*, vol. 71, pp. 1128–1142, 2023.
[49] R. Zhang, L. Cheng, S. Wang, Y. Lou, W. Wu, and D. W. K. Ng, "Tensor decomposition-based channel estimation for hybrid mmWave massive MIMO in high-mobility scenarios," *IEEE Trans. Commun.*, vol. 70, no. 9, pp. 6325–6340, Sep. 2022.
[50] L. Pucci, E. Matricardi, E. Paolini, W. Xu, and A. Giorgetti, "Performance analysis of a bistatic joint sensing and communication system," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC)*, Seoul, South Korea, May 2022, pp. 73–78.
[51] O. E. Ayach, S. Rajagopal, S. Abu-Surra, Z. Pi, and R. W. Heath Jr., "Spatially sparse precoding in millimeter wave MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 13, no. 3, pp. 1499–1513, Mar. 2014.
[52] J. J. Bussgang, "Crosscorrelation functions of amplitude-distorted Gaussian signals," Res. Lab. Electron., Massachusetts Inst. Technol., Cambridge, MA, USA, Tech. Rep. 216, Mar. 1952.
[53] J. B. Kruskal, "Three-way arrays: Rank and uniqueness of trilinear decompositions, with application to arithmetic complexity and statistics," *Linear Algebra Appl.*, vol. 18, no. 2, pp. 95–138, 1977.
[54] M. Kim and Y. H. Lee, "MSE-based hybrid RF/baseband processing for millimeter-wave communication systems in MIMO interference channels," *IEEE Trans. Veh. Technol.*, vol. 64, no. 6, pp. 2714–2720, Jun. 2015.
[55] T. Lin, J. Cong, Y. Zhu, J. Zhang, and K. Ben Letaief, "Hybrid beamforming for millimeter wave systems using the MMSE criterion," *IEEE Trans. Commun.*, vol. 67, no. 5, pp. 3693–3708, May 2019.
[56] "NR; physical channels and modulation," 3rd Generation Partnership Project (3GPP), version 18.2.0, Tech. Specification (TS) 38.211, Mar. 2024.
[57] M. N. Boroujerdi, S. Haghighatshoar, and G. Caire, "Low-complexity statistically robust precoder/detector computation for massive MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 17, no. 10, pp. 6516–6530, Oct. 2018.
[58] A. Pascual-Iserte, D. Palomar, A. Perez-Neira, and M. Lagunas, "A robust maximin approach for MIMO communications with imperfect channel state information based on convex optimization," *IEEE Trans. Signal Process.*, vol. 54, no. 1, pp. 346–360, Jan. 2006.

Luning Lin (Graduate Student Member, IEEE) received the B.E. degree in electronic science and technology from Zhejiang University, Hangzhou, China, in 2021. She is currently working toward the Ph.D. degree with the College of Information Science and Electronic Engineering, Zhejiang University, Hangzhou, China. Her research interests include array signal processing, wireless communication, and integrated sensing and communication.

Chengwei Zhou (Senior Member, IEEE) received the Ph.D. degree in electronic science and technology from Zhejiang University, Hangzhou, China, in 2018. In 2017, he was a Visiting Researcher with the University of Technology Sydney, Sydney, NSW, Australia. From 2018 to 2020, he served as a Postdoctoral Research Fellow with the State Key Laboratory of Industrial Control Technology, College of Control Science and Engineering, Zhejiang University, Hangzhou, China. Since 2020, he has been with the College of Information Science and Electronic Engineering, Zhejiang University, where he is currently a Tenure-Track Professor. His research interests are in the areas of array signal processing, direction-of-arrival estimation, and adaptive beamforming. He is currently serves as an Associate Editor for *International Journal of Communication Systems*, *Franklin Open*, and an Editorial Board Member for *Journal of Signal Processing, Journal of Information and Intelligence*, and *Journal of Radars*. He was also a Committee Member for several IEEE international conferences, including the 11th IEEE Sensor Array and Multichannel Signal Processing Workshop (IEEE SAM 2020) and the 13th IEEE/CIC International Conference on Communications in China (IEEE/CIC ICCC 2024). He was the recipient of the 2021 IEEE Signal Processing Society Young Author Best Paper Award, IEEE ICASSP 2023 Top 3% Paper Recognition, 2019 IET Communications Premium Award, and ISAP 2020 Best Paper Award.

Hang Zheng received the B.E. degree in transportation engineering from Tongji University, Shanghai, China, in 2019, and the Ph.D. degree in electronic science and technology from Zhejiang University, Hangzhou, China, in 2024. From 2022 to 2023, he was a Visiting Student with Aalto University, Espoo, Finland. His research interests include array signal processing, tensor signal processing, and AI for sensing. He was the recipient of the ISAP 2020 Best Paper Award, the ICAUS 2022 Best Paper Award, the IEEE ICASSP 2023 Top 3% Paper Recognition, and the 2023 Excellent Scientific Paper from China Association for Science and Technology.

Zhiguo Shi (Fellow, IEEE) received the B.S. and Ph.D. degrees in electronic engineering from Zhejiang University, Hangzhou, China, in 2001 and 2006, respectively. Since 2006, he has been a Faculty Member with the College of Information Science and Electronic Engineering, Zhejiang University, where he is currently a Full Professor. From 2011 to 2013, he visited the Broadband Communications Research Group, University of Waterloo, Waterloo, ON, Canada. His research interests include array signal processing, localization, and Internet-of-Things. He was the recipient of the 2019 IET Communications Premium Award, and coauthored a paper that received the 2021 IEEE Signal Processing Society Young Author Best Paper Award. He was also the recipient of the Best Paper Award from the ISAP 2020, the IEEE GLOBECOM 2019, the IEEE WCNC 2017, the IEEE/CIC ICCC 2013, and the IEEE WCNC 2013. He was the General Co-Chair of the IEEE SAM 2020 and served as an Editor for the IEEE Network. He is currently serving as an Area Editor of IEEE INTERNET OF THINGS JOURNAL, and an Associate Editor for IEEE SIGNAL PROCESSING LETTERS and IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY. He is an elected member of the Sensor Array and Multichannel (SAM) Technical Committee of the IEEE Signal Processing Society.

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.

Sergiy A. Vorobyov (Fellow, IEEE) received the M.Sc. and Ph.D. degrees in systems and control from the National University of Radio Electronics, Kharkiv, Ukraine, in 1994 and 1997, respectively. He is a Professor with the Department of Information and Communications Engineering, Aalto University, Finland. Since his graduation, he held various faculty and research positions with the University of Alberta, Canada; the Joint Research Institute between Heriot-Watt University and Edin burgh University, U.K.; Duisburg-Essen University and Darmstadt University of Technology, Germany; McMaster University, Canada; the Institute of Physical and Chemical Research, Japan; and the National University of Radio Electronics, Ukraine. His research interests include optimization and multiliner algebra methods in signal processing and data analysis; statistical and array signal processing; sparse signal processing; computational imaging; estimation, detection and learning theory and methods; and multiantenna, very large, cooperative, and cognitive systems. He was the recipient of the 2004 IEEE Signal Processing Society Best Paper Award, the 2007 Alberta Ingenuity New Faculty Award, the 2011 Carl Zeiss Award (Germany), the 2012 NSERC Discovery Accelerator Award, and the IEEE ICASSP 2023 Top 3% paper recognition, and other awards. Since 2016, he has been currently a Senior Area Editor for IEEE SIGNAL PROCESSING LETTERS. He was an Associate Editor for IEEE TRANSACTIONS ON SIGNAL PROCESSING, from 2006 to 2010, and IEEE SIGNAL PROCESSING LETTERS, from 2007 to 2009. He was a member of the Sensor Array and Multi-Channel Signal Processing and Signal Processing for Communications and Networking Technical Committees of the IEEE Signal Processing Society, from 2007 to 2012, and from 2010 to 2016, respectively. He was the Track Chair for Asilomar 2011, Pacific Grove, CA, USA; the Technical Co-Chair for IEEE CAMSAP 2011, Puerto Rico; the Tutorial Chair for ISWCS 2013, Ilmenau, Germany; the Technical Co-Chair for IEEE SAM 2018, Sheffield, U.K.; the Technical Co-Chair for IEEE CAMSAP 2023, Costa Rica; and the General Co-Chair for EUSIPCO 2023, Helsinki, Finland.

Robert W. Heath Jr. (Fellow, IEEE) is the Charles Lee Powell Chair in wireless communications with the Department of Electrical and Computer Engineering, University of California, San Diego. He is also President and CEO of MIMO Wireless Inc. From 2020 to 2023, he was the Lampe Distinguished Professor with the North Carolina State University and a Co-Founder of 6GNC. From 2002 to 2020, he was with The University of Texas at Austin, most recently as Cockrell Family Regents Chair in Engineering and Director of UT SAVES. He is the author of the book titled *Introduction to Wireless Digital Communication* (Prentice Hall, 2017) and *Digital Wireless Communication: Physical Layer Exploration Lab Using the NI USRP* (National Technology and Science Press, 2012), and co-authored of *Millimeter Wave Wireless Communications* (Prentice Hall, 2014) and *Foundations of MIMO Communication* (Cambridge University Press, 2018). He has been a co-author of a number award winning conference and journal papers including recently the 2017 Marconi Prize Paper Award, the 2019 IEEE Communications Society Stephen O. Rice Prize, the 2020 IEEE Signal Processing Society Donald G. Fink Overview Paper Award, the 2021 IEEE Vehicular Technology Society Neal Shepherd Memorial Best Propagation Paper Award, and the 2022 IEEE Vehicular Technology Society Best Vehicular Electronics Paper Award. Other notable awards include the 2017 EURASIP Technical Achievement award, the 2019 IEEE Kiyo Tomiyasu Award, the 2021 IEEE Vehicular Technology Society James Evans Avant Garde Award, and the 2025 IEEE/RSE James Clerk Maxwell Medal. In 2017, he was selected as a Fellow of the National Academy of Inventors. In 2024, he was selected as a Fellow of the American Association for the Advancement of Science. He was a Member-at-Large on the IEEE Communications Society Board-of-Governors (2020–2022) and the IEEE Signal Processing Society Board-of-Governors (2016–2018). From 2018 to 2020, he was an Editor-in-Chief of IEEE Signal Processing Magazine. He is also a licensed Amateur Radio Operator, a Private Pilot, a registered Professional Engineer in Texas. He is an elected member of the National Academy of Engineers.

Authorized licensed use limited to: University of Pisa. Downloaded on August 24,2025 at 16:28:53 UTC from IEEE Xplore. Restrictions apply.